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Half a Century of Research on Garner Interference and the Separability–Integrality Distinction

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Research in the allied domains of selective attention and perceptual independence has made great advances over the past 5 decades ensuing from the foundational ideas and research conceived by Wendell R. Garner. In particular, Garner's speeded classification paradigm has received considerable attention in psychology. The paradigm is widely used to inform research and theory in various domains of cognitive science. It was Garner who provided the consensual definition of the separable-integral partition of stimulus dimensions, delineating a set of converging operations sustaining the distinction. This distinction is a pillar of today's cognitive science. We review the key ideas, definitions, and findings along 2 paths of the evolution of Garnerian research: selective attention, with a focus on Garner interference and its relation to the Stroop effect, and divided attention, with focus on perceptual independence gauged by multivariate models of perception. The review tracks developments in a roughly chronological order. Our review is also integrative as we follow the evolution of a set of nascent ideas into the vast multifaceted enterprise that they comprise today. Finally, the review is also critical as we highlight problems, inconsistencies, and deviations from original intent in the various studies.

Keywords: Garner interference, separable–integral dimensions, similarity, models of perception and decision

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The first impression that people have merely by opening their eyes is that of separate recognizable objects in their field of view. They see faces, buildings, cars, and trees. They furthermore notice the attributes of the stimuli—the gender of the face, the color of the tree, the direction of the moving car. This seemingly effortless achievement supports the belief that the stimuli and their attributes are separate autonomous articles in the field of view, that they are independent in perception. However, such naïve beliefs are shattered when people perform the simplest of selection tasks with respect to the stimuli and their attributes. People are puzzled to learn that they cannot identify the color of the tree without noticing the tree's name, cannot decide the speed of the car without noticing its direction of motion, and cannot judge the height of the building without being affected by its width. For whole objects, too, people experience difficulty in locating a face caused by mere presence of a car (especially, if they are car experts), and it takes them longer to identify the top of the building when a tree is also present in the scene; in the concert hall, people can find attending to the string section of the orchestra arduous when another melody is played concurrently by the woodwinds. Clearly, people's conscious bids to attend selectively are often foiled by intrusions from task-irrelevant information. In this review, we explicate the concept of *selectivity* and its allied concept of perceptual *independence*, gauged through the highly influential toolkit of the Garner paradigm, Garner effects, and Garnerian theoretical notions.

By definition, each activity of everyday life requires at least some ability to attend selectively to the relevant feature in the environment, excluding irrelevant or distracting features. In the absence of this ability, one cannot concentrate on reading the newspaper in the cafeteria, negotiate the traffic when driving or walking, or conduct a conversation with a friend. Facility in isolating the task-relevant attribute is indispensable for adaptation and survival. However, as our examples also illustrate, people's attempts to attend selectively are not always successful. In this review, we discuss the conditions that promote and that disrupt the selectivity of attention. A major attendant condition for successful selectivity is the presence of perceptual independence: Selective attention to a single attribute of an object is possible if and only if perceptual independence is present across the various attributes of the same object. When discussing perceptual independence, we show how the associated models extend the reach of Garner's ideas to include tasks of categorization and attendant theory.

The selection aspect of the Garner project is naturally based on tasks of selective attention, whereas the independence aspect is mainly based on tasks of divided attention. Both aspects form integral parts of the Garner program. In the first part of the review, we follow the ways that selectivity and Garner interference have

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We have written this review with the blessed memory of Wendell R. Garner in mind - teacher, role-model, and life-long friend and inspiration. We would like to thank Devin Burns, Colin M. MacLeod, Michael Kane, James Pomerantz, Vered Shakuf, and an anonymous referee for their generous investment and comments that improved this project. Preparation of this article was supported by an Israel Science Foundation grant (ISF-274-15) to Daniel Algom.

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served to inform major substantive issues of cognitive psychology. In the second part, our attention shits to models of multivariate perception that treat the notion of independence in a large theoretical context. We conclude the review with a list of major results and problems that must be accounted for by any successful theory of the Garner effects.

The Selective Attention Path

We introduce the Garner paradigm in relation to the Stroop effect, followed by a review of the impact of Garner's measures and ideas in major domains of cognitive psychology. We then discuss the historical roots of the Garner project, and conclude with an examination of the various criticisms leveled against the paradigm and its standard interpretation.

Arguably the easiest path to the world of Garner, in particular to the classic Garner paradigm, is through reference to the older and more familiar Stroop effect (Stroop, 1935). The Stroop effect of selectivity (indeed, its failure) is the single most popular phenomenon in current cognitive science (see MacLeod, 1991 and MacLeod & MacDonald, 2000, for reviews; see also, Eidels, Townsend, & Algom, 2010). Nevertheless, some 30 years after its discovery, another phenomenon, Garner interference, has been attracting a great deal of attention (see Figure 1). Despite the explosion of work in the past decades, Garner effects are not nearly as well-known as Stroop effects. It is the goal of this review to make Garner effects and theory more accessible to the community of cognitive psychologists at large.

The Stroop Effect as a Special Case of the Garner Measure of Selectivity

The Stroop effect is the prime example of the human failure to attend selectively to an individual aspect of the stimulus. When naming the color in which color words are printed, people seem unable to ignore the meaning of the carrier words. To gauge the influence of the task-irrelevant words, the Stroop effect is defined as the difference in color naming performance between congruent (the word naming its own color, such as RED in ink color red) and incongruent (the word and its ink color conflict, such as the word RED in green) stimuli. Equation 1 defines the Stroop effect (in all experimental designs):

$$\text{Stroop effect} = \text{MRT(incongruent)} - \text{MRT(congruent)} \quad (1)$$

where MRT is mean reaction time (RT) to name the ink color. An analogous formula exists for error. Considering Equation 1, had people focused exclusively on the target dimension of ink color, there would not have emerged a content-dependent difference in color naming (= the Stroop effect). Therefore, better performance with congruent than with incongruent stimuli shows that people paid at least some attention to the task-irrelevant word, thereby compromising exclusive focusing on the target ink color.

However, it is immediately apparent that all multidimensional stimuli are not also Stroop stimuli. As a glimpse at Equation 1 shows, the effect is calculable only for stimuli that can be characterized as congruent or incongruent. Now a red triangle is a multidimensional stimulus, but it is neither more nor less congruent (or incongruent) than, say, a blue circle. The quality of congruity does not apply to stimuli composed of a shape in color. The Stroop effect is neither defined nor calculable for such objects (see Algom, Chajut, & Lev, 2004, for a discussion of the distinction between Stroop and non-Stroop objects). However, the question of selectivity is present with equal force with respect to non-Stroop stimuli, too. Can people attend selectively to color and ignore the shape of the same stimulus (Cohen, 1997)? Can a pilot focus on azimuth and momentarily ignore height while preparing for landing? Can people focus on the vertical elevation of a dot and ignore the dot's horizontal position (Garner & Felfoldy, 1970)? The answer is "yes" in the first case, but "no" in the latter two. Clearly, selective attention is consequential for adaptation and survival with Stroop and non-Stroop stimuli alike, but the Stroop effect is no longer serviceable with the latter stimuli. Another measure is therefore needed to assess selectivity. This measure is Garner interference.

The Garner measure is the product of an independently developed body of work on the nature of perceptual objects. It has long been noticed that all objects of perception are not experienced in the same way. Some objects are experienced in their totality as undivided wholes. Other objects do not form a *Gestalt*, but rather impress the observer as a composition of distinct attributes. A bright circle is readily decomposed to its constituent dimensions of luminosity and shape, but color, which is the product of three physical dimensions, can be decomposed perceptually only imperfectly and with cognitive effort (Garner & Felfoldy, 1970; Jones & Goldstone, 2013). How does one capture this difference between the two types of objects?

Of multiple attempts at dissociation, it was the system developed by Wendell R. Garner (1962, 1970, 1974a, 1974b, 1976; see also, Garner, 1991) that proved theoretically powerful and resilient to the fads and vicissitudes of psychological experimentation. The most important theoretical notion to emerge from Garner's work is the distinction between *integral* and *separable* dimensions. Gestalt objects are composed of integral dimensions that interact in processing, whereas non-Gestalt objects are composed of separable dimensions that do not interact in processing. This partition has formed a pillar of cognitive science ever since the foundations laid down in Garner's work half a century ago.

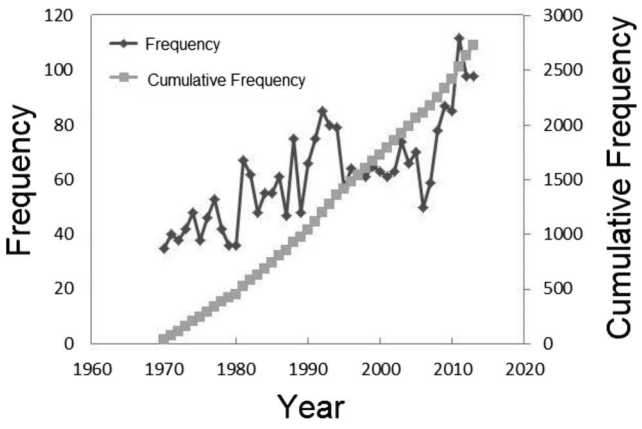


Figure 1. The annual and cumulative frequency of citations of Garner articles from 1970 to 2014. The estimates are conservative because all Garner studies do not include "Garner" in the title.

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The paradigm developed by Garner is the benchmark model for the diagnosis of separable and integral dimensions. Its trademark measure, Garner interference, provides a universal yardstick for assessing the selectivity of attention to any stimulus dimension of choice (i.e., Stroop and non-Stroop dimensions alike). The general rule is simply stated: Selective attention to a single aspect of a stimulus is possible with objects composed of separable dimensions, but selective attention is bound to fail with objects composed of integral dimensions.

Garner's Speeded Classification Paradigm

Suppose that you are presented with rectangles of different height (tall, short) and width (wide, narrow) and that your task is to classify the height of the rectangle presented on each trial while ignoring its width. Recall that rectangles do not form Stroop objects (combinations of width and height do not create congruent or incongruent stimuli), so that the Stroop effect cannot be used for assessing your success at exclusive attention to height. How does the Garner tool address this question?

The Garner logic is straightforward. In one block of trials, the experimenter keeps the task-irrelevant dimension of width constant so that only target height varies from trial-to-trial (Baseline). In another block of trials, the experimenter lets task-irrelevant width vary randomly (Filtering). The task for the participant is the same in both blocks of trials: to classify height as speedily and accurately as possible. The difference in performance between the two conditions reveals the success of selective attention to height. Poorer performance in Filtering than in Baseline would show that irrelevant variation in width took a toll on performance, compromising fully selective attention to height. Alternatively, comparable performance in the two conditions would show that irrelevant width was successfully filtered out. The parity attests to perfect selectivity of attention to the target height. The difference in RT to classify height in the two blocks defines the measure known as *Garner interference* (Pomerantz, 1983):

$$\text{Garner Interference} = \text{MRT (Filtering)} - \text{MRT (Baseline)} \quad (2)$$

where MRT is the mean latency to classify height. An analogue formula exists for error. The complete Garner paradigm includes a third condition, Correlation. Under Correlation, as in Filtering, irrelevant width also varies from trial to trial. However, under Correlation it does so in a corresponding manner with the target dimension of height. For example, on all (most) trials, a long width comes with a short height or vice versa.

Performance under Correlation is often better than that in Baseline, the difference called, *redundancy gain*. The reason is obvious: The nominally irrelevant dimension of width now becomes predictive of the relevant dimension of height. Note, though, that the presence of redundancy gain simultaneously signals the failure of fully selective attention to height. In other cases, performance under Correlation can be on a par with that in Baseline. This parity, when it is present, documents good selective performance with respect to target height.

As we document in this review, Garner's speeded classification paradigm has been applied to a large variety of perceptual dimensions. Two general patterns of performance have emerged. Some dimensions exhibit an all-inclusive uniformity in performance: Speed (and accuracy) of performance remain invariant across Baseline, Filtering, and Correlation. With such dimensions, the participants neither suffer Garner interference nor reap redundancy gain in performance. For other dimensions, performance is best under Correlation and worst in Filtering. With these latter dimensions, the participants suffer Garner interference from orthogonal variation in Filtering but reap redundancy gain from corresponding variation under Correlation.

Garner elected to characterize these patterns in terms of the partition between separable and integral dimensions. The following RT equalities define separable dimensions:

$$\text{MRT(correlation)} = \text{MRT(baseline)} = \text{MRT(filtering)} \quad (3)$$

where MRT is the mean RT to respond to the relevant dimension. By contrast, integral dimensions are defined by the following RT inequalities:

$$\text{MRT(correlation)} < \text{MRT(baseline)} < \text{MRT(filtering)} \quad (4)$$

where MRT is again the mean RT to respond to the target dimension. In Figure 2, we illustrate the Garner paradigm along with the two general patterns of results.

When proposing the separability–integrality distinction, Garner was aware of the need for converging operations to sustain this theoretical concept (Garner, Hake, & Eriksen, 1956). Therefore, task and measures of divided attention and categorization (discussed in the second part of the article) play a critical role in supporting the theoretical notion of separability–integrality. Re-

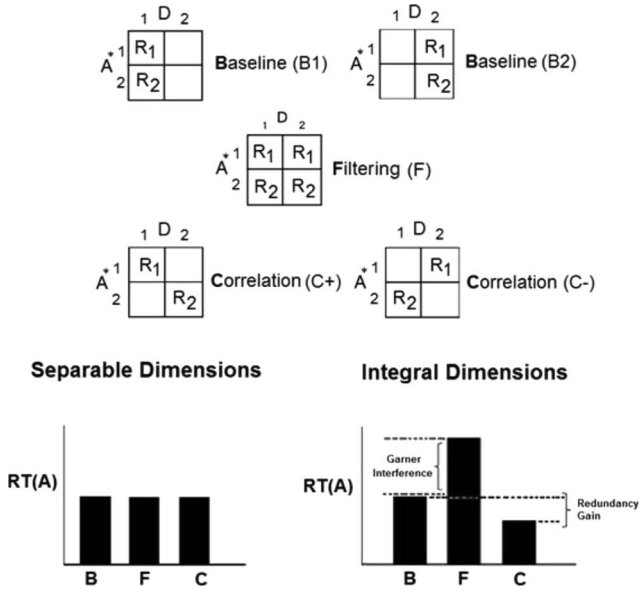


Figure 2. Schematics of Garner's speeded classification paradigm. The three basic conditions are depicted on the top. The asterisk indicates that dimension A, not dimension D, was the relevant one for responding. R1 and R2 depict the correct responses to the two values of the relevant dimension A. The Correlation conditions depict two possible ways of correspondence in the appearance over trials of the values dimensions A and D. To connect to the example of rectangles in text, dimension A is height and dimension D is width. Two prototypical outcomes, tapping integral and separable processing, are depicted at the bottom.

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turning to our example of height and width of rectangles, they proved to be integral dimensions (regardless of whether height or width was the target dimension for responding). The results exhibited both Garner interference and redundancy gain (e.g., Freud & Ganel, 2015; Ganel & Goodale, 2003; Kunde et al., 2007; Macmillan & Ornstein, 1998; Monahan & Lockhead, 1977; Tanzer, Freud, Ganel, & Avidan, 2013; see also Krantz & Tversky, 1975; Weintraub, 1971). By contrast, shape and color or shape and light, are prototypical separable dimensions (e.g., Bonnel & Prinzmetal, 1998; Cohen, 1997; Garner, 1974a, 1974b; Handelt & Imai, 1972; see also, Tanzer et al., 2013).

The Relation Between Stroop and Garner Effects

The difference in derivation granted, both measures are based on speeded classification of a single attribute of a complex object. The Stroop effect is readily derived from the Garner paradigm (when the dimensions are Stroop dimensions). Consider the Filtering condition (see Figure 2 again): One diagonal entails congruent stimuli, the other incongruent stimuli. Their difference defines a Stroop effect. Virtually all studies in the vast Stroop literature (see MacLeod, 1991) have used a design like Filtering (typically, with four values per dimension). Another Stroop effect can be derived from Baseline: Although the task-irrelevant dimension is held constant, variation of the target dimension results in congruent combinations half of the time and incongruent combinations in the remaining half. Finally, a unique, between-block Stroop effect emerges under Correlation, which we discuss below.

With Stroop dimensions, the Correlation condition divides into two meaningful blocks of trials: positive correlation (C+) and negative correlation (C−; see Figure 2, again). In the positive correlation block, all (most) trials entail congruent stimuli (e.g., the word RED appears in red and the word GREEN appears green), whereas in the negative correlation block all (most) trials entail incongruent stimuli (RED in green and GREEN in red). Robert Melara was the first investigator to explore the behavior of Stroop dimensions within the Garner design (Melara & O'Brien, 1987). Performance in the C+ block is usually better than that in the C− block, the difference marking a (between-blocks) Stroop effect. One must be aware, though, of a real problem when blocking congruent trials: People can switch (imperceptibly) to word reading instead of color naming. Often, performance in the C+ block is better than that in Baseline, indicating redundancy gain. In contrast, performance in the C− block is sometimes poorer than that at Baseline, indicating redundancy loss. Redundancy loss is usually small (if recorded at all). The reason is that the predictive power of the irrelevant word in the C− block (as strong as that in the C+ block) is offset by the semantic conflict of word and color. With non-Stroop dimensions, the particular way that one conjoins values in the correlation conditions does not make a difference.

Since the pioneering studies by Melara (1989, 1992; Melara & O'Brien, 1987; Melara & Marks, 1990a, 1990b, 1990c; Melara & Mounts, 1993), explorations of the shared and distinct empirical features of Garner and Stroop effects have resulted in a large literature (see Melara & Algom, 2003; Pomerantz, 1983, and Pomerantz, Pristach, & Carson, 1989, for reviews and discussion). In a major empirical effort, Pomerantz, Pristach, and Carson (1989) directly compared Stroop and Garner effects across 63 separate experiments conducted in their laboratory. Three features of the results were notable: First, the two effects were uncorrelated, indicating that these two measures of selectivity tap different sources; second, Garner interference was larger in each case than Stroop interference; and, third, Garner interference was never negative, whereas the Stroop effect sometimes was. We should note that Clark and Brownell (1976) were probably the first investigators to measure Stroop and Garner effects within the same experiment, although their study was aimed at other matters.

Finally, there is a compelling conceptual relation between the two effects. Garner interference registers the fact that the observer noticed the presence of task-irrelevant variation and that this variation affected performance. The Stroop effect registers the fact that the observer noticed the kind of variation and was sensitive to the momentary value of the irrelevant dimension (corresponding or conflicting with the relevant dimension). Because one cannot notice the kind of variation without noticing variation, the presence of the Stroop effect implies the presence of Garner interference, although the reverse does not hold. The stimuli need not be subjected to semantic analysis to generate Garner interference, but semantic analysis is indispensable for generating the Stroop effect. This neat logic is mostly supported (Algom et al., 1996; Melara & Mounts, 1993; Pansky & Algom, 1999, 2002; Pomerantz, 1983, 1991; Pomerantz et al., 1989; Sabri, Melara, & Algom, 2001), but not always (Eitan, Schupak, Gotler, & Marks, 2014; Martino & Marks, 2000; Patching & Quinlan, 2002; Van Leeuwen & Bakker, 1995). Concerning the latter studies, Eitan, Schupak, Gotler, and Marks (2014) noted that they form a small minority (limited mainly to cross-modal investigations).

Since its launch in the late 1960s of the last century (Garner, 1969, 1970, 1974a, 1974b, 1976; Garner & Felfoldy, 1970; Gottwald & Garner, 1972, 1975; Handelt & Imai, 1972; Hyman & Well, 1967, 1968; Imai, 1966; Imai & Garner, 1965; Lockhead, 1966, 1972; Shepard, 1964; see also, Egeth, 1967; Felfoldy, 1974), the test and the attendant concepts have been firmly established in cognitive science. Garner interference serves as the primary procedure of choice for diagnosing selectivity.

Garner Effects in Various Domains of Cognitive Science

Applications span the gamut from time perception (Zakay, Bibi, & Algom, 2014) to language (Lew et al., 1997) to emotion (Mama, Ben-Haim, & Algom, 2013) to audition (Melara & Marks, 1990a) to brain processes (Boenke et al., 2009; Caclin, McAdams, Smith, & Giard, 2008; Cant & Goodale, 2006; Cant et al., 2008; Chen & Melara, 2014; Drucker, Kerr, & Aguirre, 2009; Ganel & Goodale, 2003) to synesthesia and cross-modal processing (Eitan, Schupak, & Marks, 2008) to memory (Melara & Nairne, 1991). Below, we discuss seven domains of cognitive science in which the Garner tools have led to important theoretical advances. We selected these particular domains for two interrelated reasons. First, each represents a central area of research within cognitive psychology. Second, the application of Garner's methods in each case has contributed to the reformulation of old problems by way of suggesting new and profitable lines of investigation.

The Processing of Faces

The attraction of the human face is irresistible with artists, psychologists, and laymen alike. Nowhere in psychology has more

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talent been invested in studying a single subject. And nowhere has the Garner paradigm made a more decisive impact. We discuss three important issues.

Face Recognition

An influential model of face recognition (Bruce, 1988; Bruce & Young, 1986; Burton, Bruce, & Johnston, 1990; Haxby, Hoffman, & Gobbini, 2000) asserts that face-related information is processed via a pair of independent channels. One channel entails information needed for individual face identification. It is this route that enables people to decide whether a particular face is familiar, the person to whom the particular face belongs, and in what prior context the particular face was seen. The second channel does not process information referring to an individual face but rather to attributes shared by all human faces. They include properties such sex, emotional expression, age, or mouth shape. According to this *parallel-route hypothesis*, the identity of faces (processed in the individual-face-identification route) and the sex of faces (processed in the generic face classification route) form separable dimensions. Consequently, judgments of neither dimension should suffer interference (or reap gain) from the other dimension. The Garner paradigm provides a ready tool for testing the separability prediction of the dual-route theory.

In a pair of influential Garner studies, Ganel and Goshen-Gottstein (2002, 2004) found that the dimensions of familiarity and sex[1] as well as the dimensions of identity and expression were strongly integral dimensions, refuting the parallel route hypothesis (and supporting a version of a single-route hypothesis). Similar results were reported in further Garner studies of face recognition (Baudouin et al., 2002; Fitousi & Wenger, 2013; Kaufmann & Schweinberger, 2004; Schweinberger, Burton, & Kelly, 1999; Schweinberger & Soukup, 1998; Wang, Fu, Johnston, & Yan, 2013; see also, Bartlett, Searcy, & Abdi, 2003, and see Lander & Butcher, 2015, for a recent review). Since these studies, the issue has lost some of its prominence so that investigators often test facial dimensions without recourse to their class membership. The Garner paradigm itself is increasingly complemented by related measures of separability, supporting the mutual interaction of facial features (e.g., Fitousi & Wenger, 2013; Soto, Vucovich, Musgrave, & Ashby, 2015; Yankouskaya, Booth, & Humphreys, 2012).

More recently, the Garner paradigm has been employed to unearth the precise processing relations across the gamut of facial attributes from gender and emotional expression (Atkinson, Tipples, Burt, & Young, 2005; Le Gal & Bruce, 2002), identity and speech (Schweinberger et al., 1999; Schweinberger & Soukup, 1998), configural and featural properties (Amishav & Kimchi, 2010; Kimchi, Behrmann, Avidan, & Amishav, 2012; see also Kimchi & Amishav, 2010), and gaze and expression (Ganel, 2011; Ganel, Valyear, Goshen-Gottstein, & Goodale, 2005; Graham & LaBar, 2007).

Face Perception

The consensus that emerges from the vast literature on face perception is that faces are processed in a holistic fashion (see Richler & Gauthier, 2014, for a review and meta-analysis; but see also, Campbell, Schwartzer, & Massaro, 2001; Fitousi, 2015, 2016; Pomerantz et al., 2003). Although the concept of holistic processing is intuitive and appealing, it is actually a loaded term that has proved evasive in manifold attempts at capturing it experimentally. The minimum stipulation shared by virtually all attempts at pinpointing holistic processing seems to be that the parts of the face are less accessible than the whole face. Thus, people "suffer cross talk from irrelevant features when judging [a] probed feature" (Farah et al., 1998, p. 486). Along the same line, selective attention has long been recognized as a potent means to test the hypothesis of holistic processing (e.g., Farah et al., 1998; Richler, Cheung, & Gauthier, 2011; Richler & Gauthier, 2014; Richler et al., 2008; Richler, Palmeri, & Gauthier, 2015; Sergent, 1984; Tanaka & Farah, 1993). It is against this background that it comes as a surprise that psychology's classic tools for assessing selectivity—Stroop and Garner effects—have been underused for probing holistic face perception. In particular, these classic measures have not been deployed to test selective attention to extended regions of faces—top half versus bottom half—that form one of the most popular displays in the literature.

Nevertheless, the Garner paradigm has been deployed to test holistic processing of facial features (e.g., mouth, eyes, nose) and their spatial configuration (e.g., Amishav & Kimchi, 2010; Pomerantz et al., 2003; Wang et al., 2013; see also Bartlett et al., 2003). For example, following the Garner protocol, the participants in the Amishav and Kimchi (2010, Experiment 2) study classified the shape of the eyes once with the same mouth on the face (Baseline) and once with the shape of the mouth varying across trials (Filtering). Performance was comparable in the two conditions, so that the authors did not record Garner interference. These Garner results challenge the consensus of holistic processing, often recording separability of facial features (see Fitousi, 2015; Loftus, Oberg, & Dillon, 2004; and Tversky & Krantz, 1969, for related research that questions holistic face processing).

It is worth pausing to ponder the scarcity of Stroop and Garner studies of selective attention to the top (bottom) part of faces. The critical factor seems to be the use of unfamiliar faces in most of the pertinent research. Because the quality of congruity does not apply to parts of unfamiliar faces, studies inevitably relied on surrogates of Stroop and Garner effects. One such task is the popular composite-face matching task (e.g., Richler & Gauthier, 2014; Richler, Palmeri, & Gauthier, 2015; Rossion, 2013; see Figure 3).

Impaired Face Perception

Prosopagnosia is a condition in which face perception is impaired either due to brain damage or is impaired congenitally (in the absence of brain damage). Studies on prosopagnosic individuals revealed disrupted holistic processing of faces (e.g., Avidan, Tanzer, & Behrmann, 2011; Busigny & Rossion, 2010; Palermo et al., 2011). Do individuals with congenital prosopagnosia exhibit a deficit in holistic processing of other visual stimuli? In a recent Garner study with prosopagnosic individuals, Tanzer, Freud,

[1] Because the dimensions of sex and familiarity both are organismic or subject variables, the pertinent stimuli cannot be manipulated by the experimenter (Kantowitz, Roediger, & Elmes, 1997). As a partial remedy, Ganel and Goshen-Gottstein (2002) used multiple face stimuli for each sex and familiarity category for speeded classification of the dimensions in the Baseline and the Filtering blocks of trials.

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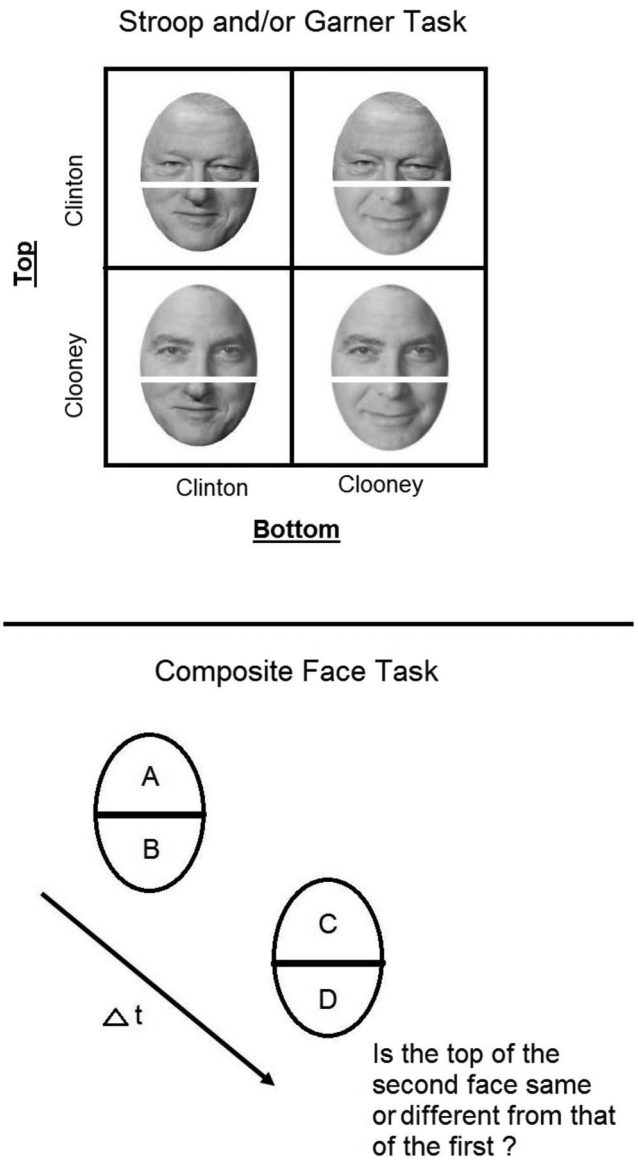


Figure 3. Top: Garner and Stroop tasks with parts (top, bottom) of *famous* faces. The "composite" faces on the minor diagonal are congruent stimuli, whereas those on the main diagonal are incongruent stimuli. The results of speeded naming of the top (or the bottom) half is eminently suitable to Garner and Stroop analysis. Bottom: the composite matching task. The task entails speeded "same–different" judgments with respect to the top (bottom) part of sequentially presented composite faces. With the unfamiliar faces typically used, the quality of congruity does not apply. The results of the composite task are not amenable to Stroop analysis. The original images were retrieved from the Internet (in the public domain): https://twitter.com/billclinton and http://www.ew.com/article/2016/01/19/george-clooney-oscars-diversity

Ganel, and Avidan (2013) did not record Garner interference for height and width of rectangles—in contrast with normal integral performance. Notably, the same patients did exhibit typical integral performance when judging dimensions of color. This latter result shows that prosopagnosics *are* capable of acting in an integral fashion. The authors concluded that congenital prosopagnosia is a *selective* deficit, affecting visual stimuli composed of shape dimensions, but not other stimuli unrelated to shape.

Numerical Cognition

The Garner model has served to challenge an accepted axiom in the domain of numerical perception. The numerical magnitude of a numeral is habitually activated just about whenever a numeral is presented for view for any purpose. For example, when people judge the *physical* size of a numeral, their judgments are affected by the task-irrelevant feature of numerical magnitude. People are faster to classify physical size as large when numerical magnitude is large than when it is small. This influence is known as the size congruity effect (e.g., Besner & Coltheart, 1979; Fitousi & Algom, 2006; Henik & Tzelgov, 1982; Tzelgov, Meyer, & Henik, 1992; Pansky & Algom, 1999). The size congruity effect is considered a potent marker of the automatic activation of alphanumerical information (Fitousi, Shaki, & Algom, 2009). If numerical magnitude is processed in an automatic fashion, then numerical magnitude and physical size should appear as integral dimensions (when people respond to physical size). This is so because the automatically processed number intrudes on (the presumably less automatically processed) physical size. However, applications of Stroop and Garner effects to the study of the size congruity effect have yielded results inconsistent with this prediction. The outcome of the Garner studies—evincing the separability of numerical and physical size—casts doubt on the maxim of automatic activation of alphanumerical information.

Employing the Garner paradigm, Algom and his colleagues (Algom, Dekel, & Pansky, 1996; Fitousi, 2014; Fitousi & Algom, 2006; Fitousi et al., 2009; Pansky & Algom, 1999, 2002; see also, Sabri, Melara, & Algom, 2001) identified several systematic biases in published studies of the size congruity effect. For example, many more values of number (typically, all the numbers in the range one to nine) than physical size (typically, small, medium, and large) were presented. Moreover, the numerical values were more discriminable than the physical values (e.g., people discriminated between four and five of the same physical size faster and more accurately than between small and large formats of the same number four). These factors rendered numerical value more salient than physical size, and, sure enough, the more salient dimension of number intruded on the less salient dimension of physical size (= the size congruity effect). When these and further factors of context were matched (Algom et al., 1996; Pansky & Algom, 1999), the dimensions appeared separable so that people did not suffer Garner interference from numerical value when responding to physical size.

Garner Interrogation of the Stroop Effect

Garner studies have helped to undermine the widespread belief in the robustness, automatic nature, and, indeed, inevitability of the Stroop effect. We already mentioned the contextual factor of dimensional salience when discussing the size congruity effect in numerical cognition. It was the seminal study of Garner and Felfoldy (1970) that revealed the foundational importance of relative dimensional discriminability. For example, they showed that circle size (large, small) and the slope of a diameter crossing the circle (steep, shallow) are separable dimensions (performance with

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neither dimension was affected by Garner interference or redundancy gain). However, once the authors made the difference in circle size more noticeable than the difference between the slopes, the former intruded on responses to the latter. Subsequently, Melara, Algom, and their associates (e.g., Algom et al., 1996; Melara & Algom, 2003; Melara & Mounts, 1993; Sabri et al., 2001) applied these insights to study the impact of differential salience on the Stroop effect itself.

The results were surprising: The magnitude, direction, and, indeed, the very presence of the Stroop effect depended on the relative salience of the word and color dimensions (Melara & Algom, 2003). When the words were more salient than the colors (the default Stroop setup in the literature), the usual Stroop effect appeared. When the dimensions were equally discriminable (by presenting appropriately pretested values), the Stroop effect sharply diminished. And, when the ink colors were made purposely more salient than the carrier words, a reverse Stroop effect emerged (by which the ink colors intruded on word reading). The experimenters were able to produce a Stroop effect, a reverse Stroop effect, or to eliminate the effect altogether at will (Melara & Mounts, 1993). Melara and Algom (2003) have culled a sample of 35 published results from the literature and examined the relation between the Stroop effect and the difference in baseline difficulty between word and color. The Pearson correlation amounted to .76, meaning that over half of the variance in published Stroop research emanates from mismatched salience at baseline between word and color. The variable of relative salience unearthed by Garner and Felfoldy (1970) thus proved of critical importance in evaluating any given Stroop result.

It is instructive in this respect to take a look at the baseline performance with each variable in the classic study of J. R. Stroop himself (Stroop, 1935). Stroop's participants named the colors of 100 squares (baseline color condition) in 63.3 s, on average, but named/read 100 words in black (baseline word condition) in 41.0 s, on average. The baseline results show that reading words was easier than naming colors when each task was performed in isolation. This difference in dimensional salience has until recently been ignored in the literature. Attention focused instead on the asymmetry in interference when the same participants performed color naming (mean of 110.3 s) and word reading (mean of 43.3 s) on 100 *combined* color-word stimuli. Donning Garnerian spectacles, however, directs one to notice the prior asymmetry at baseline. Is it possible that the Stroop effect derives not from automatic word reading or from a generic dominance of word over color, but from the simple fact that the more salient dimension intrudes on the less salient dimension more than vice versa?

The Garner interrogation of the Stroop effect led to the discovery of a further factor of context: dimensional correlation. Recall that the defining property of the different conditions within the Garner paradigm is dimensional correlation: It is zero in Filtering (and trivially zero at Baseline due to the constant value of the task-irrelevant dimension), but it is positive (or negative) under Correlation. The presence of a correlation in the last condition is conductive to good performance, redundancy gain, due to the fact that the nominally irrelevant dimension becomes predictive of the target dimension. Redundancy gain, in turn, signals the failure of fully selective attention to the target dimension. Do color and word correlate over the experimental trials in the typical Stroop task, thereby generating the Stroop effect?

To answer this question, consider the most common design used in the vast Stroop literature. Four color words (say, RED, GREEN, BLUE, and YELLOW) are combined factorially with four corresponding ink-colors (red, green, blue, and yellow) to create 16 color-word Stroop stimuli (see Figure 4). Note that in this basic matrix, four combinations (in the minor diagonal) form congruent stimuli, and the remaining 12 (off-diagonal) combinations form incongruent stimuli. The mismatch in the number of congruent and incongruent combinations in the base matrix is important when we consider another fixture of published Stroop experiments: An equal number of congruent and incongruent stimuli are presented in the experimental session. Obviously, this is possible only when each congruent stimulus is presented more often than each incongruent stimulus. However, espousing this practice, the Stroop experimenter unintentionally introduces a correlation between word and color over the experimental trials. The a priori probability of a color given a word is not equal across words. The resulting word–color correlation is fatal for selective attention. Noticing first the irrelevant word on a trial, the participant has a greater than chance probability of anticipating the target color. When the nominally irrelevant dimension is in fact correlated with the relevant dimension (the typical situation in the Stroop experiment), the participants then attend to the irrelevant dimension and thus open themselves up to Stroop interference (see Figure 4).

The power of dimensional correlation to generate or to eliminate the Stroop effect was first demonstrated in a systematic fashion by Dishon-Berkovits and Algom (2000; but see, Algom et al., 1996; Logan & Zbrodoff, 1979; Lowe & Mitterer, 1982). The authors

Word \ Color	red	blue	green	yellow
RED	9	3	3	3
BLUE	3	9	3	3
GREEN	3	3	9	3
YELLOW	3	3	3	9

Figure 4. The anatomy of the Stroop experiment: Four color-words are combined factorially with four ink colors to produce 16 colored-word combinations. The entries are frequencies of presentation in 72 trials in a typical "balanced" experiment where trials in the congruent and incongruent conditions occur with equal frequency. The four combinations on the minor diagonal are congruent stimuli, whereas the 12 off-diagonal combinations are incongruent stimuli. The only way to equate the frequency of congruent and incongruent stimuli in the experimental block—the accepted practice—is to present each congruent stimulus more often than each incongruent stimulus (in this case, three times as often). In this most common example, each congruent stimulus is presented nine times and each incongruent stimulus three times, making 36 trials with each type in all. This design creates a correlation over the experimental trials between the nominally irrelevant words and the target ink colors.

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showed that when the dimensional values are allocated in a random fashion, thereby creating zero color–word correlation, the Stroop effect largely diminishes. Recently, the study of correlation lurking within the Stroop task has gained traction, uncovering a wealth of information (e.g., Bugg, 2012, 2015; Bugg & Crump, 2012; Bugg & Hutchison, 2013; Crump, Gong, & Milliken, 2006; Jacoby, Lindsay, & Hessels, 2003; Schmidt & Besner, 2008).

Following Garner's lead, Melara and Algom (2003) have identified further contexts (see also, Sabri et al., 2001). They found (based on a sample of published studies) that the salience and correlation contexts alone have a multiple correlation of .87 with the Stroop effect. This outcome is notable: Despite the variation in populations, methods, manipulations, goals, and stimuli, these two contexts alone explain over 75% of the observed variance in the Stroop effects found in the literature. It is important to realize that the contexts are not side effects, but rather are part and parcel of the experiment. Garner's insight can be phrased as follows: By creating the stimuli and by implementing the design, the experimenter necessarily creates the various contexts and, to a considerable extent, determines the result.

Perceptual Development

The idea that there is a developmental shift from a holistic mode of object processing in young children (between 4 and 7 years of age) to a more analytic mode of processing in older children and adults has been dominating pertinent research for decades (e.g., Gibson, 1969; Kemler Nelson, 1989; Shepp, 1978, 1989; Smith & Kemler, 1977; Werner, 1957). Although the maxim has been revised and modified over the years, the basic idea has survived the challenges (Campbell et al., 2001). The Garner paradigm offers a precise formulation and a rigorous test of the assumed shift. Repeated applications of the paradigm have revealed that object perception becomes more separable with age, an outcome consistent with the *separability hypothesis* (Shepp, 1978; see also Kemler, 1983).

Applying the Garner paradigm, it has been found that young children reap redundancy gain with correlated values and suffer Garner interference with orthogonal values (the integrality supporting holism), but neither effect appears with older children and adults (the separability attesting to analytic rather than holistic perception; Shepp & Swartz, 1976; Smith & Kemler, 1978; see also, Shepp, 1989). For the sake of completeness, we should mention that the precise nature of holistic processing with young children forms a topic of debate (e.g., Schwarzer, 1997; Wilkening & Lange, 1989; see also, Anderson & Cuneo, 1978; Wolf & Algom, 1987). Therefore, Garner effects provide for an operational definition of the shift, pinpointing the age at which it occurs. The systematic deployment of Garner's tools has helped to clarify not only classification (Smith, 1981, 1983), but also further developmental aspects including conservation (Kemler, 1983) and concept learning (Kemler Nelson, 1984; Ward & Scott, 1987).

Vision for Perception Versus Vision for Action

According to a widely accepted model suggested by Goodale and Milner (1992), vision for *action* and vision for *perception* are mediated by distinct neuroanatomical streams. The dorsal pathway (that includes projections from the primary visual cortex [and the superior colliculus via the pulvinar] to the posterior parietal cortex) is said to mediate visually controlled *action*, whereas the ventral stream (from primary visual cortex to temporal lobe) is assumed to subserve conscious visual *perception* (e.g., Ganel & Goodale, 2014; Goodale, 2001; Mishkin, Ungerleider, & Macko, 1983; Ungerleider & Mishkin, 1982; Whitwell, Milner, & Goodale, 2014; but see, Franz et al., 2000; Vishton, Rea, Cutting, & Nuñez, 1999). Moreover, Goodale and Milner (1992) have suggested that the action system processes information in a selective fashion so that when we reach to grasp an object we are free from contextual influences. The perception system, by contrast, acts in a relative or holistic fashion that preserves the (metric) relations between the object and its surroundings (without requiring precise information about the absolute size of the object). This contrast between analytic and holistic modes of operation easily translates into the testable hypothesis of separable versus integral processing in action and perception.

In Garnerian language, the question of interest is this: Are the dimensions of a given object separable in grasping, but integral in perception? Applying the Garner paradigm, Ganel and Goodale (2003) had participants make speeded judgments (perception) or perform speeded grasping (action) with respect to the width (wide, narrow) of rectangles while ignoring their height. The authors recorded an appreciable Garner interference in perception, but none in action. Since this study, the Garner test has grown to be the tool of choice for investigators interested in the functional dissociation of the two major visual systems (e.g., Ganel & Goodale, 2014; Janczyk & Kunde, 2012; Kunde et al., 2007; Schum et al., 2012; but see, Hesse & Schenk, 2013). The Garner studies served as a point of departure for probing the imperviousness to context of the visuomotor system by other means (Ganel, Chajut, & Algom, 2008; Ganel et al., 2012).

The Perception of Sound

The world of sound is created by the confluence of a large set of variables. For example, tones are characterized by attributes such as loudness, pitch, timbre, duration, harmony, volume, density, or location. Timbre, for one, is composed of a subset of variables itself. Melodies and rhythms are created by changing pitch and duration in systematic ways, thereby serving as our primary means of governing traditional music (Krumhansl & Iverson, 1992). How do people perceive the various dimensions of a tone? Do the dimensions interact when hearing the sound or are the attributes processed in an independent fashion?

A large volume of Garner studies has been devoted to the examination of this question (e.g., Ben-Artzi & Marks, 1999; Grau & Kemler Nelson, 1988; Caclin, Giard, Smith, & McAdams, 2007; Caclin et al., 2008; Dyson & Quinlan, 2004, 2010; Krumhansl & Iverson, 1992; Melara & Marks, 1990a,1990b, 1990c; Mondor, Zatorre, & Terrio, 1998; Watanabe, 1988; Wood, 1975). To illustrate, in the study by Grau and Kemler Nelson (1988, Experiment 2), the participants classified tones into soft or loud with pitch held constant (Baseline), with pitch varying orthogonally from trial-to-trial (Filtering), and with pitch varying in concert with loudness (Correlation). The mean RT to classify loudness was 475 ms in Baseline, 577 ms in Filtering, and 442 ms in Correlation. The appreciable amounts of Garner interference and redundancy gains recorded supported the integrality of tone perception. The same

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stimuli were shown to interact in the same manner when pitch was the relevant dimension for responding. These results by Grau and Kemler Nelson (1988) have proven general. Auditory stimuli strike the listener as a Gestalt (Iverson, 1995): The listener cannot concentrate on one attribute without noticing (all) the other attributes of the sound.

The Psychological Reality of Impossible Objects

Garner's ideas open up avenues for future research. Consider the unique class of impossible objects—two-dimensional representations of three-dimensional objects that do not exist (or are extremely unlikely to exist) physically in actual three-dimensional space (Gregory, 1966, 1970, 1997; Penrose & Penrose, 1958; Rock, 1983; Wade & Swanston, 1991). The Dutch artist Maurits Escher famously provided several remarkable drawings of impossible objects. What is the psychological status of impossible objects? The question is far from resolved; some texts of perception ignore the theme altogether, other texts subsume it under the rubric of visual illusions (e.g., Schiffman, 2003; Wade & Swanston, 1991) and still other texts treat the subject as a logical paradox (Rock, 1983). Underlying this discontent is the uncertainty regarding the psychological reality of impossible objects. Are possible and impossible objects perceived and recognized the same way? Schacter and his associates (Schacter & Cooper, 1993, 1995; Schacter et al., 1991) argued that they are not. Because global representations of impossible objects are impossible, such objects lack a stable structure. That, in turn, impairs performance with priming and recognition (but see, Ratcliff & McKoon, 1995; Williams & Tarr, 1997).

Recent deployment of the Garner paradigm has revolutionized the field. In two Garner studies, Freud, Ganel, and Avidan (2013, 2014) found robust Garner interference for *both* possible and impossible objects. Notably, this pattern held when the participants judged a dimension of depth, the feature producing impossibility in the first place. These Garner results show that impossible objects do possess psychological reality for the observer (apparently to the same extent that possible objects do). Of course, the issue is not fully settled (Freud & Ganel, 2015), but the Garner approach and findings have provided grounds for optimism.

Interim Summary

We have surveyed the remarkable impact of the Garner paradigm in various fields of cognitive science. This influence granted, the various aspects of the paradigm have been subjected to rigorous scrutiny, unearthing several problems and potential confounds. Before we address the criticisms we give a brief sketch of the developments culminating in the conception of the Garner paradigm.

A Bit of History: The Road to Garner's Speeded Classification Paradigm

The bit, information theory's fundamental unit, has indeed played a role in paving the road to the Garner paradigm and its attendant concepts. Another influence was the early studies of RT, although measures based on speeded responses were fairly sporadic in that era. We review both of these roots of Garner's work.

Information Theory: Does Identification Improve With the Addition of Dimensions to the Stimulus?

Suppose that your task is to identify tones that differ in loudness so that when a tone is sounded you respond by giving the proper numeral. It is important to note that the task is not that of discrimination because all adjacent tones are perfectly discriminable from one another. When only three or four tones are presented, you never confuse them. However, with five or more tones, confusions become so frequent that identification no longer can be trusted. This result means that people cannot identify more than six to seven tones of different loudness without confusion. In the language of information theory, this number equals approximately 2.5 bits of transmitted information, T (Attneave, 1959; Garner, 1953, 1962, 1974b; Garner & Hake, 1951; Miller, 1956; Pollack, 1952, 1953; Shannon, 1948). This result seems far out of line with what we experience in everyday life. People can identify accurately any one of several hundred faces, any one of thousands words or songs. The likely explanation lies in the number of independent attributes that characterize the stimuli (Miller, 1956). Songs, faces, and words differ from one another in many ways, whereas the tones in our example differ from one another in only a single aspect, loudness.

The penultimate sentence is notable: Identification improves only when the added dimensions are *independent* of each other. Suppose that a person can identify without errors five lines that differ in length, so that T equals 2.4 bits of information. Suppose further that a different color is added to each line. This added dimension improves identification considerably, so that T now equals to 4.2 bits for the colored line stimuli (meaning that the person can now identify without errors 20 lines; cf. Algom, Wolf, & Bergman, 1985; Baird & Noma, 1978). Consider now a counterexample. A certain chemical presented at discriminably different intensities affords 1.8 bits of transmitted information. When these stimuli are enriched by putting them in mixtures with another chemical, recognition does *not* change so that an observer can still identify (by smelling) less than four mixtures (Berglund, Berglund, & Lindvall, 1971; see also, Algom & Cain, 1991a, 1991b; and Cain & Algom, 1997). In this latter case, transmitted information to the compound does not exceed that observed with each of the individual components. The reason is that the resulting odors feel unitary to the extent that the components are hardly discernable. Collectively, the presence of a change in T (or lack thereof) is easily understood in terms of the independence (or lack thereof) of the constituent dimensions: The tested dimensions are independent in the first example (line length and color) but not in the second example (pairs of chemicals).

Summarizing a large volume of literature, Garner (1962, p. 127) concluded that "as long as the dimensions are perceptually *independent*, then addition of new dimensions will increase total discrimination and information transmission" (emphasis added; see also Eriksen & Hake, 1955). Because the notion of independence itself lacked a rigorous definition (Garner & Lee, 1962; Garner & Morton, 1969), the closely related idea of separable dimensions gained traction. The presence of a relation between the separability of the tested dimensions and the amount of gain in T has been increasingly recognized. As our examples show, there is an appreciable gain in identification when the (added) dimensions are separable (line length and color), but there is a small or negligible

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gain when the dimensions are integral (mixtures of certain odorants). Lockhead (1966, 1970; Monahan & Lockhead, 1977) has directly demonstrated the dependence of T on the separability of the stimulus attributes by manipulating the nature of the tested dimensions.

Garner's profound insight concerned the converse of this relation and can be summarized as follows.

> *The amount of change in T can serve, in turn, as a measure of separability.*

This insight forms the foundation of the entire Garner edifice. If performance improves with the addition of a second dimension, then the dimensions are separable. If, on the other hand, the addition of another dimension leaves identification unchanged, the dimensions are integral. The measure of T obtained within an absolute identification experiment thus joins Garner interference to provide a set of converging operations sustaining the separability-integrality partition (Garner, Hake, & Eriksen, 1956; see also Baird & Noma, 1978).

Lockhead (1966) arguably was the first experimenter to apply the triumvirate—Baseline, Filtering, and Correlation—to test the relationship between line length and vertical position (with T as the dependent variable). His results documented the presence of the relation between the nature of the constituent dimensions and the transmitted information. However, the first schematic depiction of the full Garner paradigm with RT appeared in an article by Garner (Garner, 1970). Garner's was also the first article to suggest the now accepted nomenclature of the separability–integrality (Lockhead had already used the term "integral dimension," but it was Garner who complemented it with "separable dimension" and made the partition a cornerstone of his research). Shepard's (1964) suggestion of unitary versus analyzable objects and dimensions is used sporadically, but Garner's 'separable versus integral dimensions' took over the field.

The problem that thwarted a fuller development within information theory was the lack of a precise model for the notion of perceptual independence. Garner and Lee (1962) pursued several alternative notions, but the breakthrough came with the publication of the seminal article by Garner and Morton (1969) on mutual uncertainty analysis. Garner and Morton (1969) developed a model of perceptual independence based on the systematic partition of the total information or uncertainty residing in a system of variables (Garner, 1962; Garner & McGill, 1956). It is unfortunate that this foundational study has been largely ignored in subsequent research (but see Ashby & Townsend, 1986; Fitousi, 2013).

Early Reaction Time Studies: From a Detour to the Main Road Leading to Garner Interference

The year of 1954 arguably marks the beginning of the road with a pair of RT studies coming from the University of Wisconsin and appearing in consecutive issues of the *Journal of Experimental Psychology*. In the study by Gregg (1954), different groups of participants made speeded responses to the binary values of a *single* dimension of a multidimensional visual dot stimulus with the task-irrelevant dimensions held constant. The relevant task in each group remained in force in a subsequent stage when the stimulus varied from trial-to-trial on all four dimensions. For example, the participants of the group responding to horizontal position (left vs. right) did so in the face of irrelevant changes in vertical location (top vs. bottom), size (large vs. small) and brightness (bright vs. dim). The participants in each group thus ignored three bits of irrelevant information when responding to the relevant dimension (comprising one bit of information). Ignoring irrelevant information exacted a very small toll (of 5 ms or less) on RT to the relevant dimension. A possible explanation of this result is that all dimensions included in Gregg's study were separable dimensions.

In the second Wisconsin study, Archer (1954) had participants responding to several dimensions of the stimulus while ignoring other dimensions of the stimulus. The instructions to different groups of participants varied with respect to the number and identity of the relevant dimensions and the number and identity of the to-be-ignored dimensions. Response time increased as a function of relevant information load but was independent of the amount of irrelevant information. By introducing more than a single relevant dimension, Archer's study departs from subsequent developments culminating in the full Garner paradigm. Nevertheless, the lack of detrimental effects of irrelevant information is notable. Collectively, the results of these two Wisconsin studies are comparable, converging on the small interference generated by separable dimensions.

Morin, Forrin, and Archer (1961) have couched the purpose of their research in remarkably modern terms (p. 89): "to determine whether human subjects can make their behavior independent of stimulus variation which is uncorrelated with the . . . behavior to be rewarded. Specifically, can well-practiced subjects disregard irrelevant cues . . .?" In the study, numerosity (one, two) and shape (circle, square) were used as dimensions. Two of the conditions are of special interest. In one, either a single circle or a single square appeared on a trial and the observer's task was speeded classification of shape. Note that in this task, the dimension of numerosity was held constant at one. In contrast, all four combinations of numerosity and shape—single circle, single square, two circles, and two squares—appeared (in a random order) in the other condition. The task for the observer remained the same, namely, to classify shape (and ignore the number of stimuli). Morin et al. (1961) found that the RT to classify shape was the same in the two conditions. Irrelevant variation in the number of stimuli did not take a toll on performance compared with a situation in which there was not irrelevant variation. The absence of interference can be explained by the fact that shape and numerosity are separable dimensions. In a well-controlled study, Fitts and Biederman (1965) replicated these findings of Morin et al. (1961).

Another influential study by Biederman and Checkosky (1970) focused on facilitation or redundancy gain, rather than on interference. The authors compared performance across Baseline and Correlation for the dimensions of size and brightness of circles. Consider first the judgment of size. In the Baseline condition, the pairs of circles differed in size with brightness held constant. In the Correlation condition, the participants still classified circle size but now the large circle appeared bright and the small circle dim or vice versa. The participants performed in a pair of comparable conditions for judgments of brightness. The results showed that the participants reaped redundancy gain for both criterial dimensions. These results challenge the diagnostic value of redundancy gain for distinguishing between separable and integral dimensions. The dimensions of size and brightness used by Biederman and Checko-

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sky (1970) are separable, yet they yielded appreciable redundancy gain in performance.

This completes our introduction of the Garner paradigm, the review of its deployment in various domains of cognitive science, and the sketch of its historical roots (a third root, similarity, is discussed within the framework of divided attention in the second part of the article). Considering its impact, it is hardly surprising that the paradigm has attracted a great deal of attention and criticism. We next review this reaction in increasing order of severity: observations, confoundings and biases, and major challenges.

Annotations With Respect to the Classic Garner Design

We first discuss three issues that are not truly criticisms, but rather attempts at providing a broader perspective on the paradigm and the possible outcomes.

Observations

Asymmetric Interference

Given a pair of dimensions, A and D, A can be separable with respect to D although D is integral with respect to A (Garner, 1976). In other words, people neither suffer Garner interference nor reap redundancy gain when judging A, but they do both when judging D. Stroop dimensions provide a ready example of asymmetric interference (Melara & Mounts, 1993; Melara & O'Brien, 1989; Sabri et al., 2001). When responding to the ink color, people cannot ignore the carrier words so that the resulting pattern is integral with respect to the target dimension of color. However, people can ignore the ink colors successfully when they read the words so that the dimensions are separable with respect to word.

Garner (1974b) suggested that the rules of logic govern dimensional relations. The logical relation of implication states that if A exists then D also exists, but that if D exists A may or may not exist. It is this logical relation that explains asymmetric relations. Consider the study by Day and Wood (1972; see also Wood, 1975) who found that a nonlinguistic dimension interfered with the classification of a linguistic dimension but that the reverse did not hold. According to the perspective of implication, a phoneme cannot exist without a pitch but a pitch can exist without any linguistic properties. Another example would be color and form because "color exists without a form, but a form must have a color" (Garner, 1974b, p. 137). The upshot is that an asymmetric relation reflects priority in levels of processing or in mere existence.

For all its appealing simplicity, the perspective of logical hierarchy did not fare well in subsequent scrutiny. Symmetry or asymmetry is rather determined by the powerful variable of relative dimensional salience or relative baseline discriminability (Algom et al., 1996; Garner & Felfoldy, 1970; Melara & Mounts, 1993; Pansky & Algom, 1999, 2002). As we have seen, when the difficulty of discrimination with the two dimensions at Baseline is mismatched, the more salient dimension (word) interferes with performance with the less salient dimension (ink color) more than vice versa. Mismatched dimensional salience has been shown to affect the Garner and Stroop outcome in a profound way.

Extending the Garner Design

Nothing in the design precludes adding further irrelevant dimensions beyond the one usually included (Adler, 2014; Ben-Artzi & Marks, 1995, 1999; Burns, 2014; Ganel, 2011). In such cases, several Baseline, Filtering, and Correlation conditions are tested, depending on the number and identity of the irrelevant dimensions as well as on the nature (correlated, partially correlated, random) of their variation. Moreover, adding a third dimension of variation can act to change the original relation—separable or integral—that was present between the original pair of dimensions (Burns, 2014; Ganel, 2011). Concerning experimental control, adding a third dimension can help in disambiguating the sources of Garner interference (e.g., prying apart the number of stimuli and the number of varying dimensions). The Garner design can be extended in another way by adding more values per dimension and/or expanding their range (Nosofsky & Palmeri, 1997a, Experiment 3, in the so-called "stretch-filtering" condition). Huettel and Lockhead (1999) have similarly manipulated the range (hence variability), using as many as 12 values of the irrelevant dimension. As a rule, performance deteriorates and Garner interference increases with range.

Garner Effects With Configural Stimuli

The defining characteristic of configural stimuli is this: Certain combinations of values create unique formations such that the components are not perceived separately. Notably, the dimensional interaction with configural stimuli differs from that usually observed with integral stimuli (Equation 4). Consider the four pairs of stimuli created by combining the binary values of the element at the left (the parenthesis opens to the left or opens to the right) and the element on the right (the parenthesis opens to the left or opens to the right):)(,)), ((, and (). Presenting these stimuli within the framework of the Garner paradigm, Pomerantz and Garner (1973) found a unique pattern of interaction: large amount of Garner interference, but no redundancy gain. The reason is that, in each case, the constituent attributes are organized into a larger holistic unit. A natural consequence of the emergence of these simple Gestalts is that selective attention to the components becomes very difficult. In a configural structure, the constituent dimensions no longer carry psychological reality. The latter are dimensions only for the experimenter but not for the observer.

Therefore, configural dimensions can be conceived as an additional alternative to separable and integral dimensions. With these dimensions, Garner interference does not necessarily imply the failure of selective attention. Rather, it shows that the observer was sensitive to the emergent features possessed by the stimuli and that, as a result, the observer divided attention across the composite pairs of parentheses. Over the past three decades, Pomerantz and his colleagues (e.g., Pomerantz, 1983, 1991, 2003, 2006; Pomerantz & Garner, 1973; Pomerantz & Portillo, 2011; Pomerantz & Pristach, 1989; Pomerantz et al., 1989; Pomerantz & Sager, 1975; Pomerantz, Sager, & Stoever, 1977) have been elucidating the psychological features of configural superiority in various domains, including visual illusions, impossible objects, and the visual arts.

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Threats to Validity: Confounding and Bias

Lack of Response Bias or a Decision Mechanism

Perceptions are not typically observable in a direct fashion. They generally pass through a decision process that also shapes the response. This decision mechanism is missing from the original Garner paradigm. In the paradigm, the response is taken at face value, which is tantamount to assuming (implicitly) that the decision criterion remains invariant across Baseline, Filtering, and Correlation. This assumption might or might not hold under all circumstances. Evidence bearing on this question is discussed within the framework of models of multivariate perception (Ashby & Townsend, 1986; Macmillan & Creelman, 2005) that we present in the second part of the article.

Possible Confounding in the Filtering Task

Variability is the crux of the Garner model: It is absent at Baseline but present in Filtering. In standard Garnerian theory, the difference in performance between Baseline and Filtering (= Garner interference) is attributed to this variation. However, the variation comes yoked together with three changes that threaten the validity of the Garner measure. First, the number of distinct stimuli is four in Filtering but only two at Baseline. A necessary (implicit) assumption of the Garner design is that perception of an attribute does not depend on the number of stimuli in the experimental block. Ashby and Maddox (1994) show that Garner interference is valid only when the assumption holds, that is, in the absence of such context effects. Second, the difference in the number of stimuli between Baseline and Filtering permits a difference in the observers' strategy. The observer is capable of adopting a change-detection strategy at Baseline so that the response is always based on the previous trial. If the stimulus remains the same, so does the response, but if the stimulus changes, the observer merely switches to the other response (Burns, 2014; Felfoldy, 1974; Nosofsky & Palmeri, 1997a). This change-detection strategy is not serviceable in Filtering, because the stimulus can change without a concomitant change in response.

The third threat is that the variability in Filtering increases uncertainty. Although the relevant dimension varies from trial-to-trial in the same fashion at Baseline and in Filtering, the variation of the task-irrelevant dimension in Filtering produces a greater amount of trial-to-trial change in the stimulus. Consequently, Dyson and Quinlan, (2010), following the method of an earlier study by Huettel and Lockhead (1999), decomposed Garner interference into two sources: macro level (between-block variability, the main cause of Garner interference) and micro level (greater trial-to-trial uncertainty in Filtering than at Baseline). Dyson and Quinlan concluded that both sources contribute to the observed Garner interference.

The method used by Huettel and Lockhead (1999) and by Dyson and Quinlan (2010) focuses on two-trial sequences: The relevant as well as the irrelevant stimulus can change (C) or repeat (R), which creates four possible sequences in Filtering: R-R, R-C, C-R, and C-C (with the first letter denoting the relevant dimension, the second the irrelevant dimension). Note that only the first and the third sequence can happen in Baseline (and only the first and the fourth are presented in Correlation). Based on these sequences, Dyson and Quinlan (2010) defined nine estimators tapping the effects associated the relevant dimension, the irrelevant dimension, and the differences across the three conditions in the Garner paradigm. Again, Dyson, and Quinlan (2010) concluded that both the greater uncertainty in Filtering as well as greater variation (the traditional account, captured by the change in the irrelevant dimension across tasks) contribute to Garner interference.

The conclusions drawn by Huettel and Lockhead (1999) were more radical: Garner interference is completely accounted for by local sequential effects, so that the term "Filtering" should no longer be used. These conclusions are too extreme for at least three reasons. First, the authors manipulated the number of values and range, which made the interpretation of sequential effects a bit more complex. Second, the authors espouse a version of Lockhead's (1972, 1979) "perceptual blob" theory, which has been convincingly refuted by Melara (1992; Melara & Marks, 1990a; see the next subsection). Third, the experiments by Huettel and Lockhead (1999) and by Dyson and Quinlan (2010) all concerned (various forms of) integral dimensions. However, the reality of separable dimensions itself militates against any substantial effect of those local confounding variables: They are present in the same form with separable and integral dimensions, yet Garner interference obtains only with the latter. The potential confounds granted, they do not modify the standard interpretation of Garner effects and the integrality–separability partition in a substantive way.

Major Challenges

Are There Integral Dimensions?

Robert Pachella and his colleagues (Cheng & Pachella, 1984; Pachella, Somers, & Hardzinski, 1981) have questioned the psychological reality of the entire class of integral dimensions. They argued that dimensional interactions called as "integral" arise as a result of the lack of correspondence between (a) the nominal dimensions imposed by the experimenter and (b) the psychologically meaningful dimensions perceived by the observer. Consider the left-hand panel of Figure 5. The stimuli were created by varying the length of the pair of straight lines originating at a right angle (e.g., edges of incomplete triangles). These dimensions do not seem to correspond to psychological attributes (cf. Cheng &

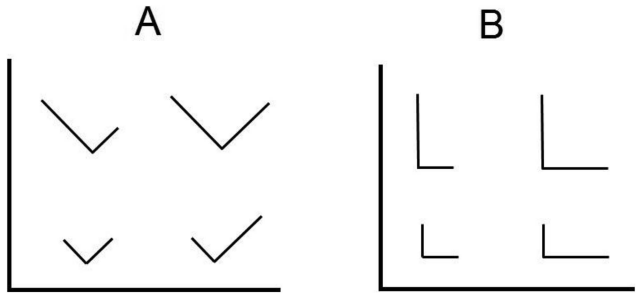


Figure 5. The panel on the left illustrates a "nonpsychological" set of stimuli, whereas the panel on the right illustrates a "psychological" set, arrived at in the present case by a 45 degree rotation. See text for explanation.

Pachella, 1984; Grau & Kemler Nelson, 1988). Note also that the two stimuli that lie on the minor diagonal share shape—a telling clue for the failure of separability. This means the presence of Garner interference when subjecting the stimulus ensemble to speeded classification along the left-hand or the right-hand line as the relevant dimension. However, Garner interference arises simply because the line task mandates the partition of stimuli in a way that violates their intuitive psychological structure (Grau & Kemler Nelson, 1988). Psychologically meaningful attributes can be obtained by rotating the axes. The results of such a rotation (by 45 degrees) are shown in the right-hand panel of Figure 5. The stimuli now vary along the dimensions of vertical and horizontal extent, which *are* psychologically meaningful. Predictably enough, these dimensions are separable (Townsend, Hu, & Ashby, 1980).

Therefore, dimensional relations fall into one of two mutually exclusive classes. There exist either separable dimensions, in which case neither dimension intrudes one on the other, or psychologically misguided dimensions, in which case they intrude on one another because they are not meaningful subjectively. The upshot is, the notion of integrality is a myth wrought about by a lack of psycho-physical correspondence (Grau & Kemler Nelson, 1988).

This extreme challenge to the separability–integrality dichotomy has not, however, stood up well in subsequent scrutiny (Foard & Nelson, 1984; Grau & Kemler Nelson, 1988; Melara & Marks, 1990a, 1990b, 1990c; Melara, Marks, & Lesko, 1992). Investigators demonstrated that dimensions that satisfied Cheng and Pachella's (1984) strict criterion for psychologically meaningful dimensions did nonetheless evince integrality in processing. Nevertheless, a more moderate principle, *the criterion of privileged axes* (Foard & Nelson, 1984), provided another criterion to identify integral stimuli as well as potentially devastating evidence regarding the spatial representation of integral stimuli. We discuss this criterion next.

The Identification of Integral Stimuli: Rotating the Reference Axes That Define the Dimensional Values

Consider the four stimuli shown in Panel A of Figure 6. They were created by two levels of sound intensity (Low: 1, 2; High: 3, 4) and two levels of frequency (Low: 1, 4; High: 2, 3). The *psychological* dimensions of loudness and pitch naturally align with these physical dimensions and can be subjected to the standard Garner paradigm of testing whether loudness and pitch are separable or integral dimensions. Now suppose that the axes in Panel A of Figure 6 are made of rigid material and that the stimulus points are also rigidly fixed to them, much like the wooden frame and interior painting of the picture on your wall. Suppose further that the rigid axes and stimulus locations of Panel A of Figure 6 are rotated by 30° (Figure 6, Panel B). This is the same situation that one sometimes experiences when hanging a new picture on the wall: The orientation is not quite straight such that the frame is misaligned with the vertical. If we wish to subject the four stimuli (now with new values of intensity and frequency) to a test by the Garner paradigm, novel names are needed to label the psychological dimensions. They can be labeled as "Apples" and "Bananas" (Melara & Marks, 1990b), indeed any convenient name will serve. Does the rotated set produce Garner interference? If it does, is Garner interference the same under the two orientations?

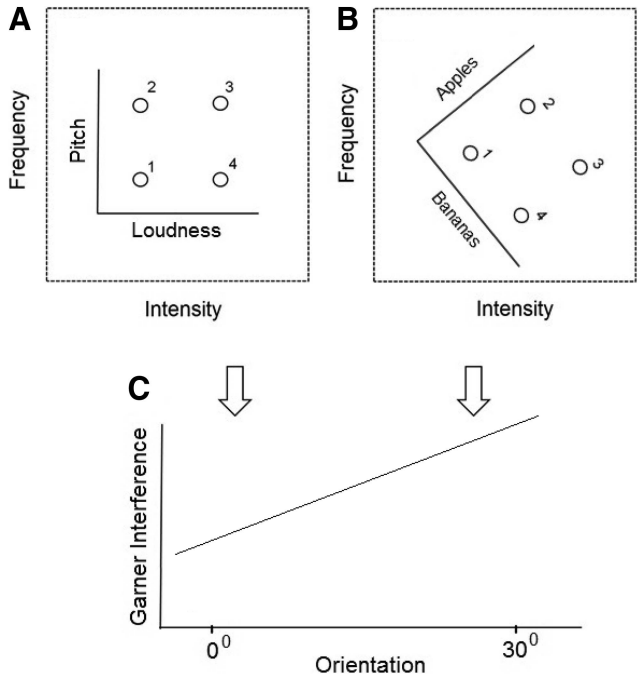


Figure 6. Privileged orientation with integral stimuli. Panel A: The psychological dimensions of loudness and pitch are aligned with the physical dimensions of intensity and frequency. As is shown in Panel C (at the left), these dimensions produce Garner interference, documenting the fact that they are integral dimensions. Panel B: The psychological dimensions are rotated (clockwise) by 30 degrees, and the new dimensions are called apples and bananas. As is shown in Panel C (at the right), when the Garner test is performed with respect to the dimensions of apples and bananas, a much larger amount of Garner interference results. Clearly, the unrotated axes are privileged psychologically, yielding the least amount of Garner interference.

To answer these questions, Smith and Kemler (1978) presented a set of colors created by the factorial combination of saturation and brightness, much like Panel A of Figure 6. They then constructed another set that corresponded to a 45° rotation of saturation and brightness, much like Panel B of Figure 6. Smith and Kemler (1978) found comparable amounts of Garner interference at the two orientations, supporting the hypothesis that, for integral dimensions, no one orientation of axes is primary. However, subsequent research using the rotation heuristics (Foard & Nelson, 1984; Grau & Kemler Nelson, 1988; Melara & Marks, 1990b; see also Melara, 1992) revealed that this conclusion was premature. In those later studies, all orientations of axes produced appreciable amounts of Garner interference, yet a privileged orientation emerged in each case (typically that aligning with the physical dimensions). Garner interference was minimal (yet still present) at the privileged orientation and it got larger with rotation of the axes (Figure 6, Panel C). The finding of a privileged orientation with integral dimensions is inconsistent with early Garner theory (see Figure 6).

The point initially made by Smith and Kemler (1977, 1978) was that, for *integral* stimuli, no pair of reference axes is primary or privileged in processing. Indeed, Garner maintained that, "psychologically, if dimensions are integral, they are not really perceived

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as dimensions at all" (1974b, p. 119). Rather, observers perceive a unified whole or a "perceptual blob" (Garner, 1981; Lockhead, 1972, 1979; Smith & Kemler, 1977, 1978). On the other hand, if the stimuli are composed of separable dimensions, the reference axes *do* comprise a defining feature of perception because such stimuli are processed in terms of their constituent dimensions. The *privileged-axes criterion* captures this difference between integral and separable stimuli: A privileged orientation should *not* be found with integral stimuli, but it should be present with separable stimuli.

Clearly, the cumulative data are inconsistent with the privileged-axes idea because integral stimuli, too, evince a privileged orientation. Moreover, the results are inconsistent with any theory assuming that integral stimuli are perceived initially as undivided wholes or as "perceptual blobs." Instead, the results show that the dimensional structure is instantaneously accessible with separable and integral stimuli alike (Jones & Goldstone, 2013; Melara, 1992; Melara & Marks, 1990b). The failure of the privileged criterion idea, namely that performance with integral stimuli does *not* remain invariant across rotation, entails ramification for the geometric representation of such stimuli, the subject of the section on similarity.

Synopsis of the Selective Attention Path and the Need for Cross-Validation

Surveying the landscape of Garner's selective attention project, the picture that emerges is that of a thriving endeavor characterized by a rapid expansion of applications to problems in central domains of cognitive psychology. We have also confronted a number of key questions and challenges that did not eventually make a serious dent in the remarkable success of the technology as a whole. However, we would be remiss at this point to ignore the growing literature using tasks of divided attention for two interrelated reasons. First, this literature comprises a large portion of current Garner research directed at the separability–integrality contrast. This literature also focuses on the complementary issue of independence in perception. Second, the results of the Garner paradigm, in particular the measure of Garner interference, cannot remain the sole arbiter of the separability–integrality distinction. It is at this point that the tasks of divided-attention come to play a critical role. When the results of divided attention are considered together with those of selective attention, circular reasoning is avoided. This double-pronged attack permits much stronger convergence of evidence supporting the integrality–separability distinction than is possible with either one alone.

The Divided Attention Path

We first discuss how studies of the psychology of similarity can be used to support the separability–integrality distinction. We also mention problems with this notion. We then review a similarity-based theory of categorization with an eye on modeling performance in the Garner paradigm. Eschewing similarity in the final sections, we discuss three models of multimodal perception that enhanced our understanding of Garnerian separability, the notion of independence, and, consequently, of the separability–integrality distinction. Integrating a decision component into the Garner paradigm and the specter of several species of separability (and integrality) comprise but two important results in modern studies along the divided attention path.

The Psychology of Similarity and the Separability–Integrality Distinction

How do people perceive the similarity between objects? In particular, is there a difference in this respect between objects composed of separable dimensions (say, a red circle and a green triangle) and objects composed of integral dimensions (say, two color patches of different brightness and hue)? Cumulative research shows that there is, and the difference is illustrated in Figure 7. The geometric representation is calculated by the method of *multidimensional scaling* (Carroll & Arabie, 1998; Carroll & Chang, 1970; Carroll & Wish, 1974) in which a set of similarity judgments are transformed to create a psychological space such that the distances between pairs of stimuli in the space are related (as closely as possible) to the actual judgments of similarity. It turns out that the best-fitting type of distance differs for separable and for integral stimuli (see Figure 7).

For objects composed of separable dimensions, similarity is derived simply by summing the differences along the constituent dimensions. For example, the similarity between the red circle square and the green triangle (Panel A, Figure 7) is determined by adding their differences along the color and the shape dimensions. This type of similarity is defined by the so called "City-block" metric (in which similarity equals the simple sum of the differences along the referent dimensions). The term "City-block" comes from the way that one traverses the distance between two points in a grid-like place like Manhattan (walking up an avenue, then turning right or left to a street).

For objects composed of integral dimensions, similarity is not based on a simple, list-wide summation of the differences along the constituent dimensions. Instead, each dimensional difference is first squared and it is these squared differences that are added together; in a final step, the square root of the sum is derived to derive similarity (Panel B, Figure 7). This determination of the distance between two points is readily recognized as the definition of distance in Euclidian geometry (specified by the well-known Pythagorean Theorem).

A single formula is capable of accommodating the two definitions of distance as special cases. This model, known as the Minkowski family of distance functions, is actually a generalization of Euclidian distance. It expresses the distance between a pair of stimuli as a power transform of their respective differences along the constituent dimensions:

$$d_{ij} = \left[\sum_{m=1}^{M} \left| x_{im} - x_{jm} \right|^{r} \right]^{1/r} \tag{5}$$

in which d_{ij} is the distance between Stimulus i and Stimulus j, x_{im} is the coordinate of Stimulus i on Dimension m, x_{jm} is the coordinate of Stimulus j on Dimension m, and M is the number of dimensions of difference. The parameter of interest is r, which determines whether the distance is City-block or Euclidian. For separable dimensions, $r = 1$, so that Equation 5 reduces to the City-block metric, whereas for integral dimensions, $r = 2$, so that Equation 5 reduces to the Pythagorean Theorem of the Euclidian metric. A voluminous literature shows that, when analyzing similarity spaces with the aid of Equation

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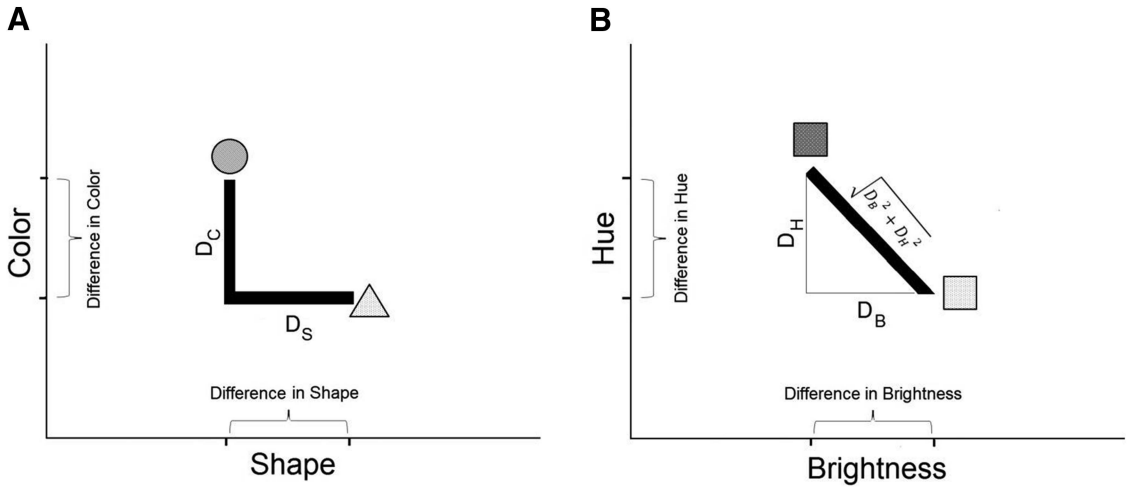


Figure 7. Different specifications of distance depicted by the bold lines. Panel A: The City-block metric. The distance (d) between the red circle and the green triangle is given by their difference on color (D_C) plus their difference on shape (D_S), so that $d = D_C + D_S$. Panel B: Euclidian metric: The distance between the pair of color patches is given by squaring their difference on hue (D_H^2) plus squaring their difference on brightness (D_B^2) and then deriving the square root of the sum, that is, $d = (D_H^2 + D_B^2)^{1/2}$, which is recognized as Pythagoras' theorem. A very simple formula can accommodate the two types of distance: $d = (D_H{}^r + D_B{}^r)^{1/r}$. When $r = 1$ in this formula, the distance is City-block and when $r = 2$ the distance is Euclidian. Equation 5 is a generalization of this formula for more than two dimensions.

5, the perceived similarity between separable stimuli is best fitted by the City-block metric, whereas that between integral stimuli by the Euclidian metric (e.g., Garner, 1974a, 1974b, Garner & Felfoldy, 1970; Maddox, 1992; Shepp, 1989).

Evolution of the Association Between the Perception of Similarity and the Separability–Integrality Distinction

In a pioneering study, Attneave (1950) asked people to judge the similarity between pairs of simple stimuli such as squares varying in size, tilt, or brightness. He found that the judgments of similarity with his simple stimuli were best accounted for by the City-block model. However, Attneave (1950) recognized that this might not hold for other kinds of stimuli or dimensions. His notion was taken up by Torgerson (1958) who suggested that the City-block metric best fits stimuli constructed of dimensions that are compelling to visual inspection (such as shapes in color or a circle and the tilt of its diameter). By contrast, dimensions of Gestalt stimuli such as those of faces or of colors are best fit by the Euclidian space. Referring to the earlier work by Attneave (1950) and Torgerson (1958), Shepard (1964) proposed a distinction between two kinds of stimuli, "those that are related as homogenous, unitary wholes, and those that tend to be analyzed into perceptually distinct components or properties" (p. 80).

Of the community of psychologists interested in the pertinent issues, it was Garner who was most impressed with these results and made them a foundation of the separability–integrality distinction. The results with judgments of similarity provided Garner with the requisite piece of converging evidence supporting that distinction. Consequently, Garner and his associates defined integral dimensions operationally as those which "lead to a Euclidian metric in direct distance scaling" (Garner & Felfoldy, 1970, p. 238; see also, Garner, 1970, 1974a, 1974b). Conversely, nonintegral or separable stimuli "yield a City-block metric in direct-distance similarity scaling" (Monahan & Lockhead, 1977, p. 95; see also, Lockhead, 1972). The upshot is that the best-fitting definition of distance joins Garner interference (and information theory's T) in providing converging support for the theoretical construct of the separability–integrality contrast.

Problems With Similarity Conceived as Distance

The crucial feat taken by Attneave (1950) was considering similarity as distance—in a rigorous mathematical sense of the term. This feat yielded rich theoretical reward, establishing the City-block-Euclidian distinction (through Equation 5). However, the notion of similarity as distance comes with a price tag: Distance in mathematics must satisfy several axioms, all of which are shown to be violated in human perception of similarity (Tversky, 1977; see also Pothos, Busemeyer, & Trueblood, 2013; Tversky & Gati, 1982). The following four axioms apply when a number assigned to a pair of stimuli is a distance. According to the condition of *symmetry*, the distance between a pair of stimuli does not depend on order (i.e., the distance from a to b is the same as that from b to a). According to the condition of *positivity*, distances cannot be negative. The axiom of *minimality* states that the distance between a stimulus and itself is minimum or zero, hence also equal for all stimuli. Finally, according to the condition of *triangle inequality*, the sum of distances among three stimuli (a to b and b to c) cannot be smaller than the distance between any pair of the stimuli (a to c). Tversky and Gati (1982) presented another condition, *corner inequality*, which is closely related to triangle inequality and is readily testable experimentally (see, Beals, Krantz, & Tversky, 1968, for full axiomatization). Again, Tversky (1977)

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has demonstrated that virtually all these distance axioms (and other conditions associated with the Minkowski metric, e.g., Suppes, Krantz, Luce, & Tversky, 1989; Tversky & Krantz, 1970; see also, Beals & Krantz, 1967; Maddox, 1992) are violated in human data.

Consider some classic examples. Ratings of similarity by humans are asymmetric, violating the symmetry axiom. People often perceive a different stimulus to be more similar to a given stimulus than the given stimulus onto itself, violating of the minimality axiom. Compelling examples (some dating to the contributions of William James; James, 1890) illustrate violations of the triangle- and corner-inequality. Consequently, "there is ample evidence disconfirming the validity of the Minkowski-r metric as a model of perceived similarity" (Maddox, 1992, p. 165).

An early attempt at salvaging the spatial model by Krumhansl (1978) was only partially successful. In another attempt, Nosofsky (1991a) suggested that similarity per se *is* symmetric and that the observed asymmetries in the data are accounted for by bias. Weighted and general multidimensional scaling models (see the next section) are indeed capable of predicting violations of the various axioms (but they require more parameters to fit the data; cf. Nosofsky, 1992; Perrin, 1992). Nevertheless, reviewing 40 years of similarity research, Pothos, Busemeyer, and Trueblood (2013) concluded that the Tversky challenge retains its force (so that their novel quantum model of similarity is not constrained by distance axioms).

We conclude this section by referring to a problem discussed earlier, the different amounts of Garner interference recorded across various orientations of the axes for *integral stimuli* (see Figure 6 again). This result is incompatible with the Euclidian metric that underlies perception of integral stimuli because in that metric the distance between any two points remains invariant regardless of the orientation of the reference axes. This anomaly alone suffices to cast doubt on the validity of spatial representations of similarity data, and, as a result, on their ability to inform the separability–integrality contrast (Melara, 1992; see also, Maddox, 1992).

We note though that, despite the problems, spatial representations retain their "beguiling appeal" (Maddox, 1992, p. 165) for students of similarity and categorization.

Similarity-Based Categorization: An Exemplar-Based Random Walk Model

In the past two decades, models of similarity-based categorization have been developed, arguably reflecting on the effects observed in the classic Garner paradigm. We review one influential theory, Nosofsky and Palmeri's (1997a, 2015) *exemplar-based random walk* (EBRW) model. In the typical task, the to-be-categorized objects are created by the combinations of values from two stimulus dimensions. For example, the stimulus set may include all 16 combinations of four values of circle size and four values of orientation of the diameter (cf. Nosofsky, 1985). For categorization, this stimulus ensemble is partitioned into two groups (defined by a critical value of circle size and a critical value of diameter orientation). Presented with a stimulus, the participant comes up with the name of the category in which the stimulus was most likely a member. For identification, the participants make a unique response to each of the 16 of the individual stimuli. The data are arranged in a "confusion matrix" that lists the frequencies of the responses for each stimulus; the correct responses are located on the diagonal, whereas the confusions lie off-diagonal. It is important to note already at this point that this categorization (or identification) task entails divided attention: The participant must attend to *both* dimensions in order to produce the response. This contrasts with the selective attention situation in the Garner paradigm. In that task, attending to a *single* dimension suffices to produce the response. We discuss more problems with attempts at applying the EBRW to the Garner milieu after presenting the model itself.

In the EBRW (as in Nosofosky's previous, *general context model*, Nosofsky, 1985, 1986, 1991a, 1991b, 1992), it is assumed that people categorize objects by storing all of the individual exemplars in memory. When a stimulus object is presented, all of the individual exemplars from memory are activated with intensity proportional to the exemplar's similarity with the stimulus. The retrieved exemplar that wins the race enters into a dynamic process of momentary category-choice that ends when a decision on the response is reached. The EBRW model, too, employs a geometrical representation, so that the exemplars are represented as points in a psychological space. The distance in the space between any two objects is derived based on direct judgments of similarity or on confusions of categorization (or identification) taken as a measure of proximity. Nosofsky and Palmeri (1997a) have also made use of the Minkowski distance formula (Equation 5), but they added a weight (and a discrimination) parameter such that:

$$d_{ij} = c\left[\sum_{m=1}^{M} W_m \left|x_{im} - x_{jm}\right|^r\right]^{1/r} \tag{6}$$

where d_{ij} is distance as in Equation 5, W_{m} ($0 \leq W_m \leq 1$) is the attentional weight that the observer assigns to Dimension m, and c is a sensitivity parameter that represents discriminability. Task dependent variation in the dimensional weights in Equation 6 enables the experimenter to model performance in a range of categorization and identification situations.

Notably, the distances in the space, d_{ij}, are *not* considered similarities in the EBRW (despite the fact that these distances are derived based direct judgments of similarity). They are rather scaled exponentially into "similarities," η, so that,

$$\eta_{ij} = e^{-d_{ij}^p} \tag{7}$$

with the exponent p determining the form of the nonlinear function. In the final step, these "similarities" are fed into the formula for predicting categorization. The conditional probability that Stimulus i, S_i, is associated with Category j, R_j, is given by,

$$P(R_J|S_i) = \frac{b_J \sum_{j \in C_J} N_j \eta_{ij}}{\sum_K b_k \sum_{j \in C_J} N_k \eta_{ik}} \tag{8}$$

where b_j is Category J's response bias, N_j is a stimulus-strength term, and η_{ij} is the similarity between Stimuli i and j. Equation 8 states that the likelihood of placing a stimulus object in a category is determined by its similarity to each stimulus in that category relative to the stimulus' similarity with each of the other stimuli in the entire ensemble (cf. Medin & Schaffer, 1978).

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Application to Garner's Speeded Classification Paradigm

According to the EBRW, redundancy gain (better performance under Correlation than in Baseline) is explained by the greater dissimilarity of the stimuli in the Correlation condition. This greater dissimilarity generates less competition in the decision process. Garner interference (longer RTs in Filtering than at Baseline) is explained by the greater number of identical memory exemplars at Baseline as compared with that in Filtering. Further insights and extensions of the basic paradigm are offered (Nosofsky & Palmeri, 1997a; see also, Burns, 2014, 2016).

The first attempt at applying the EBRW to model Garner effects (Nosofsky & Palmeri, 1997a, Experiment 3) included collecting data from the typical conditions of Baseline, Filtering, and Correlation (along with several further conditions). The EBRW predictions faithfully reproduced the order of the observed group means. Nevertheless, one should note several points. First, application of the EBRW mandates performing *two* experiments: categorization (the to-be-modeled task) and ratings of similarity. The latter is needed to derive the vital similarity parameters. However, ratings of similarity and the attendant multidimensional scaling were *not* performed in this Garner study. Instead, the similarity parameters were 'borrowed' from other experiments (or set in an arbitrary fashion). Second, the attentional weights, too, were set in an arbitrary fashion. Third, unlike the categorization experiments included in the same article, modeling in the Garner case did not entail (a) fitting of RTs to each individual stimulus, (b) fitting of RTs to each individual observer, as well as (c) predictions with respect to changes in RT with practice. Many of these problems are addressed in a recent study by Little, Wang, and Nosofsky (2016). A much earlier attempt (Nosofsky, 1985) yielded anomalous results. It is amply documented that the circle-radius stimuli used in that study are composed of separable dimensions with performance captured by the City-block metric (e.g., Garner, 1974b; Garner & Felfoldy, 1970; Kadlec & Hicks, 1998; Melara, Marks, & Potts, 1993; Shepard, 1964). However, when Nosofsky (1985) applied his earlier general context model to identification of such stimuli, he found support for the Euclidian metric.

Despite the fresh insights and the good fit to sets of empirical data, the application of the EBRW to the Garner paradigm is plagued by problems. The first problem is that the model in unable to distinguish between objects composed of separable and of integral dimensions—the heart of the Garner enterprise. If the model applies to integral stimuli (Nosofsky & Palmeri, 1997a, 2015), then, obviously it cannot decide between the two types of stimuli. A second problem is more fundamental. The primary focus of the EBRW is whole *objects* rather than *dimensions* of objects (although objects are still defined as an integration of their dimensions). It is whole objects, rather than single attributes or dimensional values, that are stored in memory (and that are then compared with the presented object). In sharp contrast, the primary focus of the Garner project is *dimensions* of objects, not whole objects. The tested dimensions often are (widely) separated in space and/or impinge on different sensory surface so that they do not form an object in any consensual definition of the term. Consider the cross-modal Garner study by Melara and Marks (1990a): The participant was sitting in one corner of the room facing a screen with a dot (at top/bottom position), while a loudspeaker in the other corner sounded a tone (of high/low pitch). Clearly, the dimensions did not coalesce into a "perceptual blob"—the sound and the visual dot each was readily perceptible at the outset. Despite the large spatial separation and the different modalities, these proved integral dimensions! In fact, a sizable portion of Garner research does not involve objects at all (e.g., Fitousi & Algom, 2006; Fitousi, Shaki, & Algom, 2009; Garner & Felfoldy, 1970; Melara & Marks, 1990a; Pansky & Algom, 2002; see also, Choplin & Logan, 2005; Garner, 1974a,1974b; Kahneman & Chajczyk, 1983; Melara & Algom, 2003).[2]

We conclude that the EBRW (as well as the earlier general context model) is not optimally suited to inform performance in the Garner paradigm.

This completes our discussion of similarity, whether as a tool augmenting the separability–integrality distinction or as a tool sustaining categorization-based modeling of Garner effects. In the following sections, we discuss models of identification and categorization that do not appeal to the notion of similarity. An important model, *general recognition theory* (GRT; Ashby & Townsend, 1986), enhanced our understanding of the Garnerian notion of separability.

General Recognition Theory: A New Look at Garner's Paradigm and Distinction

In the standard GRT experiment, the participant identifies the stimulus on *all* the pertinent dimensions. The set of stimuli is created by the factorial combination of values of two dimensions. If the dimensions are shape (square, triangle) and color (red, green), then the stimulus set includes a red square, a green square, a red triangle, and a green triangle. These four stimuli are presented many times and the task for the observer is to report the value of the stimulus on *both* dimensions. A principled comparison of the rates of accuracy (error) can yield valuable information on the separability of the dimensions of color and shape.

[2] There are further problems. A fourth problem concerns the nonlinear scaling of distance in the space (Equation 7). People judge directly the similarity between the presented stimuli and the respective distances in the space faithfully reflect their perceptions (as attested by the excellent values of goodness of fit). Now distorting nonlinearly people's judgments of similarity and calling that result "similarity" is dubious. The nonlinear transformation might be warranted when the space is derived based on confusions of categorization (cf. Luce, 1959, 1963; Shepard, 1987), but the exponential scaling is not convincing when the observers make direct similarity ratings between the stimuli. A fifth and closely related problem is that the EBRW considers *confusions* of categorization (or identification) and direct judgments of *similarity* synonymously (see, e.g., Nosofsky, 1991a, p. 97). However, confusability and similarity are distinct variables (Melara, 1992). Confusability is a performance measure. People attempt to avoid incorrect categorizations. Similarity is a perceptual or phenomenal measure. People do not attempt to avoid (dis)similarity; they simply perceive it. Erroneous similarity does not exist. Most important, the two variables behave differently as a function of stimulus difference (depending on range). The EBRW model, as well as other theories of similarity (Ashby & Perrin, 1988; Smith, 1989), should respect the fact that confusability is conceptually distinct from similarity (Melara, 1992). A sixth and final complication is associated with the attention weights, namely, that they cannot be assigned with certainty in the Correlation condition. As Ashby and Maddox (1994) point out, an observer can base her responses on the *task-irrelevant* dimension to produce the same level of performance as that of an observer attending the task-relevant dimension.

One can compare the accuracy of *color* identification (red vs. green) once for the trials in which the shape was a square and once for the remaining trials in which the shape was a triangle. If red and green are discriminated at the same level with square and with triangle (i.e., regardless of shape), then the *perceptual separability* of color with respect to shape is indicated. The parity means that shape did not make a difference in color discrimination. One can perform the same analysis with respect to identification of shape (square vs. triangle) once for red trials and once for green trials. Again, comparable performance would indicate the presence of perceptual separability of shape with respect to color.

The same accuracy data can be exploited to derive another measure of color discrimination—the value used by the observer to *decide* whether the color was red or green. This measure is based on the idea that the observer adopts a fixed criterion such that if the percept falls on one side of the criterion the observer responds "red" and that if the percept falls on the other side the observer responds "green." Again, the criterion can be calculated twice, once for color decision with square as the shape and once for color decision with triangle as the shape. If the values of the criterion remain the same, then the dimensions of color and shape satisfy *decisional separability*. The same calculation can be made for decision of shape across the two values of color. Comparable criteria would indicate the presence of decisional separability of shape with respect to color.

It is apparent that the derivation of perceptual and decisional separability follows the procedures of signal detection theory (Green & Swets, 1966; Macmillan & Creelman, 2005). Specifying perceptual separability for color can be done by comparing the pair d' values derived for red versus green, once with square and once with triangle as shape. Specifying decisional separability is similarly accomplished by comparing the respective pair of β. The theory entails additional measures and a network of theorems (see Kadlec & Townsend, 1992, for computational details, theorems, and proofs; see, Ashby & Soto, 2015, for a recent review). In Figure 8, we present an example of the perceptual space posited by the theory: The relative locations of the stimuli are determined by the observer's identification performance, so that the space provides a convenient geometric representation of perceptual and decisional separability (or their lack thereof).

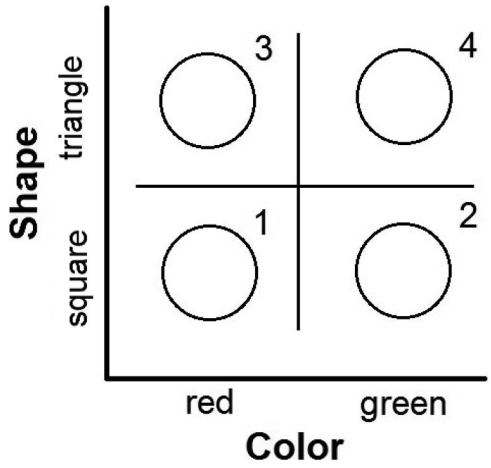


Figure 8. Illustration of general recognition theory's perceptual space. The circles within the space represent the four colored shapes (numbered 1–4). In this example, the horizontal distance from 1 to 2 is the same as that from 3 to 4, which shows that color identification or discrimination is unaffected by shape. Color thus is perceptually separable with respect to shape. The same parity exists for the vertical distances 1–3 and 2–4, showing that shape, too, is perceptually separable with respect color. The linear decision bounds are parallel to the two axes, documenting the presence of decisional separability for both dimensions, The circular shapes (equal probability contours cut from four bivariate Gaussian distributions) satisfy a condition called *perceptual independence* (not discussed in text along with further measures and theorems of the theory).

Application to Garner Research

The introduction of decisional separability posed a challenge to the uniqueness of Garnerian separability. Applying GRT, a given pair of dimensions can be perceptually and decisionally separable, be the first but not the second, be the second but not the first, or be neither (Ashby & Maddox, 1990). Only the first of the four possibilities satisfies the empirical separability used in Garner research. Dimensions can be diagnosed as integral not because perceptual separability has failed but because the observers elected to use an integral decision strategy.

In two early studies (Ashby & Gott, 1988; Ashby & Maddox, 1990), the participant categorized stimuli composed of separable dimensions (the stimuli were drawn from two experimentally generated distributions). Nevertheless, Ashby and his colleagues found that the observers used an *integral* decision rule (to maximize accuracy). Consequently, Ashby and Maddox (1990) suggested that perceptual separability is a property of the stimulus but that decisional separability is at the observer's discretion. People will use an optimal decision rule even when it means integrating information from perceptually separable dimensions. One must be a bit circumspect, though, given that the Ashby studies included merely two prototypical stimuli and that those stimuli entailed a correlation between the dimensional values. Another subtle problem is that the separable dimensions used were not themselves tested in the Ashby studies, but were drawn from the general Garner literature. However, it is this literature that is criticized by Ashby and his colleagues for lack of uniqueness.

Are Perceptual Separability Within GRT and Garnerian Separability Comparable?

Close scrutiny shows that they are not. The fundamental reason is that, unlike the Garner paradigm, the GRT experiment does not include a condition in which one of the dimensions is held constant. There are further subtle differences (see Algom, 2016, for a detailed analysis). Both Garner separability and perceptual separability tap the ability of an observer to exempt target performance from influence by a second dimension, but they do so by different means. The few empirical studies that derived both Garner and GRT measures (Algom & Edelstein, 1998; Fitousi, 2013, 2014; Fitousi & Wenger, 2013; Kadouri-Labin, 2006) yielded inconclusive results. Garner interference and failure of perceptual separability tended to occur in tandem, but did not always, and the individual variability was large.

In the remainder of this discussion, we briefly address two GRT developments of consequence to the Garner project.

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Speeded General Recognition Theory

Ashby and Maddox (1994; see also Ashby, 1989, 2000; Maddox & Ashby, 1996; Townsend, Houpt, & Silbert, 2012) developed an RT version of the theory. The Ashby extension is based on the RT-distance hypothesis by which RT decreases as the distance between the stimulus' perceptual effect and the decision bound increases (see Algom, 1992; Marks & Algom, 1998; Moyer, 1973; Moyer & Landauer, 1967, on the role of a distance effect in psychophysics). With the distance hypothesis in force, the speeded model reproduces the standard Garner effects in the Baseline, Filtering, and the Correlation conditions (see Equation 2 and Theorems 2–6 in Ashby & Maddox, 1994). We note, though, that Nosofsky and Palmeri (1997b) found the EBRW to yield a better fit to the data of their Garner experiment than did the Ashby and Maddox (1994) model.

A rigorous analysis of the Garner paradigm based on the speeded model (Ashby & Maddox, 1994) showed that the Garner tests of Filtering and Correlation are imperfect. For perceptually separable dimensions, Filtering will correctly reveal the absence of Garner interference (i.e., comparable means across Filtering and Baseline), but Filtering can be insensitive when certain types of integrality are present in the data (e.g., mean-shift integrality, to be discussed next). The problems with respect to the Correlation task are more serious. A redundancy gain may occur (a) because the dimensions are truly integral *or* (b) because the dimensions are truly separable, but the observer adopted optimal (i.e., integral) responding.

Forms of Integrality

Inspecting the GRT perceptual space (see Figure 8), it is clear that there can be many different forms of integrality (Maddox, 1992). For example, integrality can be present even under equal interstimulus distances in the perceptual space, if, for example, increasing the level of B tends to increase the level of A (see Figure 9). This particular form is called *mean-shift integrality* (assuming that the marginal distributions have the same variance). The reverse is also possible, so that only the variances differ, *variance-shift integrality*. Of course, both the means and the variances may vary. The species of mean-shift integrality, in particular, has been subjected to extensive analysis (Ashby & Maddox, 1994; Kingston, & Macmillan, 1995; Kingston, Macmillan, Dickey, Thorburn, & Bartels, 1997; Maddox, 1992). The reality of multiple forms of integrality has led some investigators to question the uniqueness of many GRT solutions (Silbert & Thomas, 2013; see also, Mack et al., 2011). For example, when linear bounds do not satisfy decisional separability, there always exists an alternative model that does, so that decisional separability is not testable. We note, though, that Ashby and Soto (2015) have recently concluded that the Silbert and Thomas challenge is "not catastrophic" (Ashby & Soto, 2015, p. 18; see also Fitousi, 2014), suggesting ways of meeting the challenge.

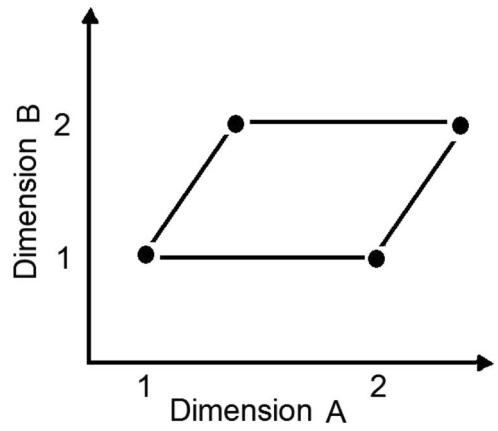


Figure 9. An example of mean-shift integrality.

We next consider an increasingly influential theory, *system factorial technology* (SFT), which speaks to a foundational issue at the heart of the Garner project: the presence of independence in processing various stimulus attributes (Townsend & Nozawa, 1995; see Algom et al., 2015, and Algom, Fitousi, & Eidels, 2016, for recent reviews and discussions).[3] Independence is the bridge that brings together the results from Garner studies of selective and divided attention: It is possible to attend *selectively* to a single attribute of the stimulus if and only if there is perceptual *independence* across the various attributes of the same stimulus (Algom, 2016).

System Factorial Technology

The task for the observer is to detect the presence of a target in the stimulus display. Suppose that the displays comprise the four Stroop stimuli, RED in red (with the former signifying the word and the latter the ink color), RED in green, GREEN in red, and GREEN in green, and that the target is the signal "red" whether in the word or in the ink color. Then, the observer is to say YES when "RED in red," "RED in green," or "GREEN in red" is presented, but to say NO when "GREEN in green" is presented. The experiment is conveniently captured by the factorial design, word-target (present, absent) × color-target (present, absent). This design contains displays with two targets (RED in red), with one target (RED in green, GREEN in red) and with no target presented (GREEN in green).

The critical question of independence is resolved by the presence and magnitude of a redundant target effect, namely, the presence of better performance with double-target displays (RED in red) than with single-target displays (RED in green or GREEN in red). The natural explanation for the fast responding to RED in red is that the word and the color targets interact in processing: The activation from both channels combines to swiftly satisfy the response criterion. However, fast responses to RED in red can also be observed when the word and color targets are processed completely independently. If the response on each trial is determined by the winner of an independent race (between color and word), then, the *mean* RTs will be faster on double-than on single-target trials (if the RT distributions for word and color overlap; Miller,

[3] The SFT addresses the question of independence in processing in a fundamental way, harnessing state-of-the-art tools of stochastic modeling (Townsend, 1974; Townsend & Ashby, 1983; see also Dzhafarov, 2003; Kujala & Dzhafarov, 2008; Schweickert, 1978; Schweickert, Fisher, & Sung, 2012; Sternberg, 1969; Townsend & Schweickert, 1989). Our exposition here is highly selective.

1982; Raab, 1962). Miller (1978, 1982, 1986) suggested a now well-known RT inequality in order to decide between independent race and integral or coactivation models,

$$Fwc(t) \leq F_W(t) + F_C(t), \quad (11)$$

in which $Fwc(t)$ is the cumulative distribution function for the "RED in red" trials and $F_W(t)$ and $F_C(t)$ are those for "RED in green" and "GREEN in red" trials. The inequality states that the cumulative distribution function for the redundant target trials, $F_{WC}(t)$, cannot exceed the *sum* of the single target distribution functions at any time t – *if* word and color are independent in processing (see Figure 10). Consequently, violations of the inequality are usually taken to reject *separable* or independent race models in favor of *integral* or coactivation models (see, Luce, 1986; Townsend & Honey, 2007, or Townsend & Wenger, 2004, for assumptions underlying the Miller inequality).[4]

A popular extension of the SFT, the *double factorial paradigm*, includes a second manipulation: Each of the words and the colors presented can appear in a highly salient or in a less-than-highly salient format. Focusing our attention only on the double-target presentations, "RED in red," the manipulation of salience creates the second factorial design: RED (in salient or less salient font) × red (in salient or less salient hue). The results of this factorial arrangement, uncovering architecture, are also diagnostic with respect to separable versus integral processing. When the results of the SFT experiment satisfy Miller's inequality and the architecture is consistent with a parallel-race (see Figure 10), the dimensions are likely processed in a separable or independent fashion. In contrast, violation of Miller's inequality in tandem with an interactive architecture would suggest integral processing (see Figure 11).

Application to Garner Research

In a recent study, Fitousi and Wenger (2013) had the same participants respond to facial features in three separate experiments—Garner, GRT, and SFT. The collective results attested to a fair, though not full, degree of convergence among related constructs in the three paradigms. In another attempt, the aforementioned experiment with Stroop stimuli was actually performed by Eidels, Townsend, and Algom (2010). The results showed that the responses were faster to displays containing redundant targets (both the word target and the color target are present) than to displays containing a single target (only the word target or the color target is present) even in cases in which the two targets were defined in a way that did not form Stroop-congruent stimuli. Stroop-congruent redundant targets (e.g., "RED in red") did not yield faster responses than did Stroop-incongruent redundant targets (e.g., "GREEN in red"). Notably, the redundancy gains observed did not derive from an interaction in the processing of the word and the color targets. The Miller inequality was always satisfied, supporting an *independent parallel race* of color and word.

These results invite the reexamination of the traditional conceptions of Stroop and Garner effects. For Stroop effects, Eidels et al. (2010) have shown that the behavioral Stroop effect *is* possible under a parallel independent race architecture (within a restricted range of parameters). Eidels (2012) has further developed this independent channels idea into a new revolutionary theory of the Stroop effect. Notably, a Stroop effect can ensue under an architecture in which the color horse does not know the position, rate, indeed the very existence of the word horse! The results of Eidels et al. (2010) pose a challenge to Garner's concepts, too. After all, Stroop dimensions are integral dimensions, which are supposed to interact in processing. If they do not always, one must search for further subtle sources for the separability–integrality distinction. Another challenge to this distinction is discussed in the next and final section.

Structured Categorization: A Synthesis of EBRW, GRT, and SFT

In standard categorization experiments, the stimuli within each category do not form any particular configuration apart from sharing category membership. In an ingenious feat Fific, Nosofsky, and Townsend (2008) modified this feature by arranging the stimuli in the category of interest in a factorial design (of the values of the constituent dimensions). The participant's decision that the presented stimulus belongs in this category is equivalent to a YES response in the SFT experiment, so that the SFT can be applied to the speeded categorization responses. The design is illustrated in Figure 12.

Fific et al. (2008) measured speeded categorization in their novel design once for stimuli composed of separable dimensions and once for stimuli composed of integral dimensions. The results revealed different architectures. For separable stimuli, the dimensions were processed in a serial fashion—the processing of a dimension awaited completion of the processing of the other dimension—and the response was given only after all information was accrued (this exhaustive strategy is actually mandated by the AND rule defining the target category, see Figure 12 again). For integral stimuli, by contrast, the processing was parallel and coactive.

Building on the foundations of the Fific et al. (2008) study, the goal of subsequent models was to provide a processing account for the observed differences between separable and integral stimuli (Fific, Little, & Nosofsky, 2010; Little, Nosofsky, & Denton, 2011; Little, Nosofsky, Donkin, & Denton, 2013; see also, Fific, 2014; Nosofsky, Little, Donkin, & Fific, 2011). Notable are the "logical-rule-based" models of categorization (Little et al., 2011; Little et al., 2013), which integrate the hitherto separate approaches of Townsend (GRT, SFT) and Nosofsky (EBRW), a synthesis originally conceived by Fific (2006). According to the

[4] The Miller inequality is recast within SFT as a statement about the capacity of the system. If the processing of a target is slowed down when presented along with another target the system is of *limited capacity* (i.e., there is not a redundant-target advantage). When the processing of a target is the same in single and in joint presentations, the system is of *unlimited capacity*. When each target is processed better in joint presentation, the system is of *supercapacity*. It was suggested (Ashby & Townsend, 1986) that capacity is unlimited with separable dimensions, but capacity is limited with integral dimensions. The absence of Garner interference with separable dimensions reflects the property of unlimited capacity that performance with an individual stimulus does not depend on the total number of the stimuli to be processed (Townsend & Ashby, 1983). Capacity analysis can also be applied to performance under the Correlation condition (Burns, 2014). Capacity produced Garner effects is an exciting idea, but it is fraught with problems and Burns (2014) recent attempt yielded inconclusive results.

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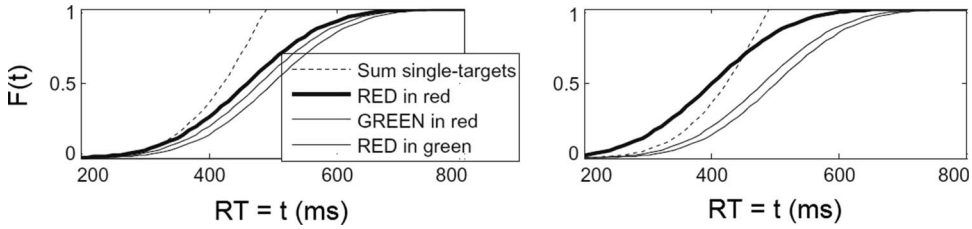


Figure 10. Different predictions for separable versus integral processing based on Miller's inequality. The results on the left satisfy the inequality because the RT observed for double targets (RED in red) does not exceed in any time the sum of the RTs for single targets. These results support separability. The results on the right, by contrast, violate the inequality because the RTs observed for double exceed the sum of the RTs observed for single targets (at least some of the time, *t*).

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unified model, for objects composed of separable dimensions people first make independent decisions about values along each dimension, with the decision along each dimension governed by a random walk process (borrowed from EBRW). The separate outcomes are then combined by the applicable logical rule—AND for the target category (because both target values are necessary) and OR for the contrast category (one or the other target is necessary). The independent processes along each dimension can be accomplished serially or in a parallel fashion. If the latter, categorization speed for the target category is determined by both processes (given the AND rule). For the OR rule in the contrast category, categorization can be determined by the random walk that finishes first (when the observer responds by a minimum time strategy) or by both processes (when the observer awaits completion of both processes in an exhaustive fashion).

For integral stimuli, by contrast, it is assumed that there is a single random walk based on the pooled RT distribution over the constituent dimensions. The reason is that observers combine values from the individual dimensions at an early stage and it is this pooled information that drives categorization. The results obtained in the studies by Little et al. (2011, 2013) reproduced the original Fific et al. (2008) pattern, evincing a coactive model for integral dimensions but not for separable dimensions. Little et al. (2013) found further that the precise mode of processing for separable stimuli depends on the spatial separation of the stimulus components. When the components were widely separated in space, the processing was serial. When the components somewhat (though not fully) overlapped, categorization was accomplished by a mixture of serial and parallel processing.

On a critical note, the assumption of single-channel processing for integral stimuli is problematic. As we recounted, investigators of the Garner program have already shown that all versions of an early "perceptual blob" (Lockhead, 1972) are untenable (e.g., Foard & Nelson, 1984; Goldstone, 1998; Goldstone & Styvers, 2001; Grau & Kemler Nelson, 1988; Jones & Goldstone, 2013; Melara, 1992; Melara & Marks, 1990a, 1990b, 1990c). Then, no independent definition or measurement is provided for the putatively separable and integral stimuli selected for study. Finally, Little et al. (2013) draw a tenuous parallel between separable dimensions, on the one hand, and separation of the dimensions in space, on the other hand. However, separation in space is not a defining property of separable processing (e.g., Fitousi & Algom, 2006; Garner, 1974a; Kahneman, & Chajczyk, 1983; Melara & Marks, 1990b; Shepp, 1989).

Theoretical Dilemma: Are There Different Species of Separability and Integrality?

The collective results of the studies based on the Fific et al. (2008) paradigm document a separability-integrality contrast that differs from that of the original Garnerian definition. Consider separability: According to Garner's definition (Equation 3), separability exists when the observer successfully ignores (variation along) the task-irrelevant dimension. Separability in this sense is trivially compromised in the categorization studies by virtue of the fact that *categorization* requires attention to both dimensions. There is no relevant (or irrelevant) dimension: All of the dimensions are relevant. Nevertheless, a systematic difference *was* found between separable and integral stimuli. How does one reconcile the two sets of definitions and findings? Are there two (or more) kinds of separability (and integrality)? There probably are. A tentative resolution would pit top-down separability (in speeded categorization) against data driven separability (in speeded classification). The problem underscores the importance of task: speeded *identification* or *categorization* of multiple attributes in SFT or EBRW versus speeded *classification* of a single attribute in the Garner paradigm. Indeed, Maddox and Ashby (1996) found that a change in task (from identification to classification) wrought about a qualitative change in representation (see also, Ashby & Lee, 1991).

Conclusion

An essential partition between two large classes of stimuli, Gestalt or holistic versus non-Gestalt or analytic, has long been suspected, but it was Garner (along with contemporary investigators such as Posner, 1964 and Shepard, 1964) who provided a firm conceptual foundation and experimental tools to sustain the division. The procedure and measure that has come to bear Garner's name, Garner interference, now rivals Stroop interference as psychology's standard measure of the selectivity of attention. His early foray into methods of divided attention, in particular, into the domain of similarity, has revolutionized the field and inspired the development of models of categorization elucidating the notion of independence and of the separability–integrality contrast. Garner's paradigm and ideas have withstood the fads and vicissitudes of half a century of research, growing into a pillar of modern cognitive science. This review documents the continuous impact that makes Garner one of psychology's most influential scientists.

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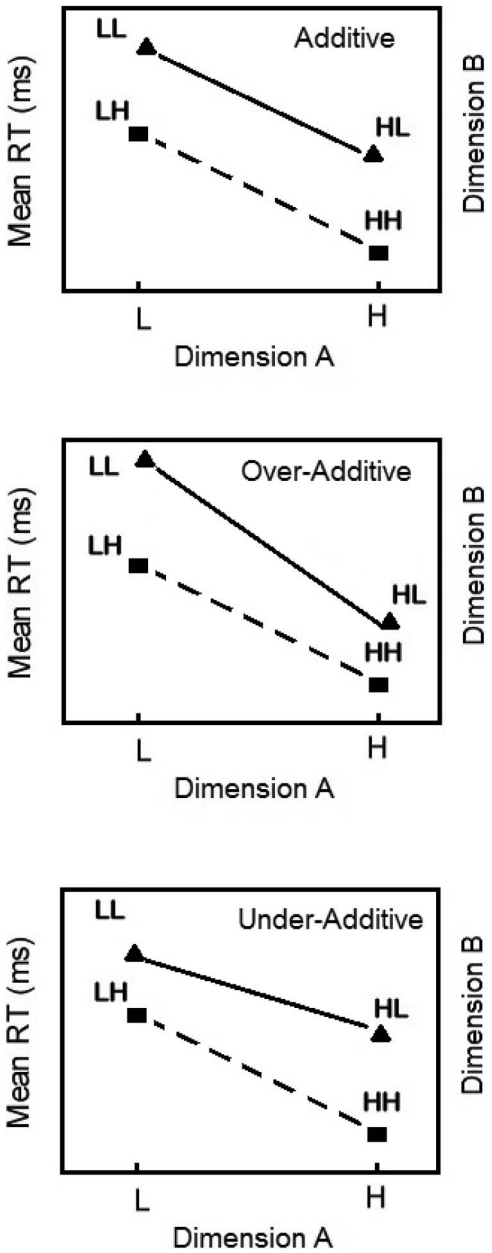


Figure 11. Three factorial plots of the mean RTs, creating additive, overadditive, and underadditive structures. If the two targets are processed in parallel, with the observer responding immediately upon minimum evidence, the plot will be overadditive. The response in this case will be always fast except when both targets are of low salience. If the observer responds only after all information has been accrued, the plot will be underadditive. In this case, the responses will be always slow except when both targets are salient. When the observer processes one target after the other in a serial fashion, an additive plot results regardless of whether the observer responds by a minimum time or by an exhaustive rule.

Although psychology remains in debt to this extraordinary scholar—Garner's contributions to psychophysics, information theory, and the psychology of the arts (and baseball!) are insufficiently recognized—there is an enormous interest in his findings and ideas concerning attention and independence.

Taking a glimpse at the title of the article, we will hazard the inevitable comparison with the excellent review of the Stroop phenomenon by Colin M. MacLeod published a quarter of a century ago on these very pages. We strived to realize the lofty goals set by MacLeod (1991, p. 163): "To survey [the field] . . ., showing critical weaknesses . . . and argue for a . . . framework . . . informed by contemporary cognitive psychology" so that "future research might profitably proceed." Hopefully, we have achieved at least some of these goals. What is clear is that the phenomena associated with Garner's ideas will continue to fascinate theorists and to challenge experimenters for years to come. We conclude by a more concrete look at this future.

A Dozen Results and Problems That Must be Accounted for by Any Successful Theory of Garner Effects

1. The heart of the Garner paradigm is irrelevant variation. However, increased variation comes yoked with an increase in the number of stimuli, an increase in trial-to-trial uncertainty, expanded sequential effects, and possible changes in the performer's strategy. Any adequate theory of Garner interference must be able to identify and isolate these sources that rest on the presence of irrelevant variation.

1a. Concerning sequential effects, trials n and $n + 1$ can entail the same or a different response (= same or different value of the target attribute), the same or a different value of the irrelevant attribute, as well as various combinations of the two. Note that four combi-

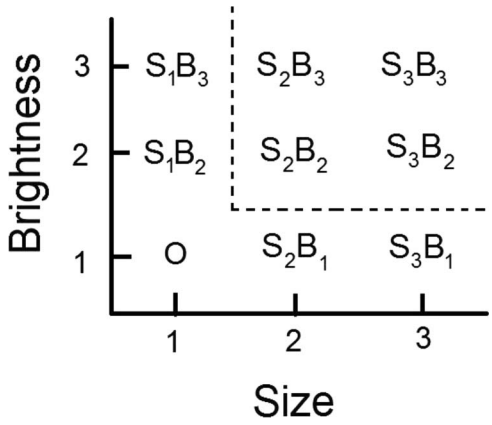


Figure 12. Illustration of the categorization structure used by Fific et al. (2008). A square is presented so that one attribute, brightness (B), can appear at three levels of intensity, dim (B1), moderate (B2), or bright (B3), and that another attribute, size (S), can similarly appear at three levels of magnitude, small (S1), intermediate (S2), or large (S3). Brightness and size are combined in a factorial fashion, with each brightness-size combination presented for speeded *categorization* according to the following rule: A square belongs to Category A, the target category, if it has a sufficiently intense brightness, $B \geq B2$, AND if it has a sufficiently large size, $S \geq S2$. Otherwise, the square belongs to Category B, the contrast category. The main feature to notice is that the four stimuli in the target category form a neat factorial design, so that these data are amenable to SFT analysis.

nations are present in Filtering, but only two at Baseline and in Correlation.

1b. Concerning uncertainty, it is greater at Filtering than in Baseline, given the larger repertoire of stimulus combinations in the former condition.

1c. Concerning strategy, the observer can adopt a change-detection strategy in Baseline by which any change in the stimulus signals switching to the other response. In Filtering, changes in the stimulus do not necessarily signal a change in the response.

1d. Nevertheless, the various effects are present in the same way testing separable dimensions, yet Garner interference is absent. The implication is that mere presence of irrelevant variation *is* the decisive factor.

2. What is the reason for the optimal performance in Baseline? Is the facility attributable to the absence of task-irrelevant variation or to the absence of irrelevant information, namely, to the fact that the value of the task-irrelevant dimension on trial n completely predicts its value on trial $n + 1$?

3. What is the influence of decision in shaping Garner's effects? This question is most critical with respect to redundancy gain in Correlation: Does the gain reflect true integrality or a change in decision (in which case the dimensions are separable rather than integral)?

4. What is the precise nature of the relation between Stroop and Garner effects? Why is it the case that the logical implication connecting the two effects is sometimes violated in empirical research? And, why do those violations occur mainly in cross-modal studies?

5. Is there a single, task-independent notion of integrality? Integral dimensions in the classic design are defined by the failure of selective attention to any one dimension. Integrality in this sense is present in a trivial sense in studies of categorization because the response mandates paying attention to all dimensions. And yet, different patterns still emerge in categorization for integral and separable stimuli. Is there a bottom-up (selective attention) and a top-down (divided attention) integrality?

6. Is there a relation between Miller's inequality (and the allied notion of capacity, see Note 4 above) and Garner interference? Does the violation of the inequality comes in tandem with Garner interference (i.e., with integral dimensions), whereas satisfying it appears with separable dimensions?

7. What is the precise relation between the apparently parallel concepts of separability and independence in the different paradigms of Garner, GRT, and SFT?

8. Is there a relation between holistic perception and integrality? In particular, if the dimensional structure is equally accessible with separable and integral stimuli, then how is holism generated?

9. Gestalt stimuli, integral stimuli, and capacity: Gestalt stimuli are composed of integral dimensions that exhibit Garner interference in processing and so are of limited capacity (see again Note 4 above). However, the same stimuli often exhibit superior performance in related tasks—characterized by supercapacity. So, are (components of) Gestalt stimuli processed supercapacity or limited capacity?

10. The influence of task: One should distinguish between speeded *classification* of a single attribute and speeded *identification* or *categorization* of multiple attributes (which compose the object). Modelers should respect the distinction and refrain from designating all tasks as "classification."

11. Similarity versus confusability: These two measures—judgments of similarity and rates of confusions—are not the same and should not be considered as comparable measures of "proximity." Modelers should respect this difference.

12. The contextual factor of relative dimensional discriminability or salience can influence the Garner effects. Needed is an independent measure of this factor, one that lies outside of the paradigm itself.

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